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Bayesian model inference with complex posteriors: Exponential-impact-informed Bayesian Quadrature

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ABSTRACT

Estimation of multimodal and sharp posteriors with nonlinear dependencies as well as the associated model evidence remains a critical challenge in many Bayesian model inference tasks such as model parameter calibration, model selection and model averaging. Bayesian Quadrature (BQ) based on approximating the logarithm of likelihood with a Gaussian Process surrogate model has been proven to be a promising scheme for multimodal inference, but the mechanism behind it has not yet been sufficiently investigated. By exploring the mechanism of exponential impact behind this, I first answer the questions “why it works?”, as well as “can it work better, and how?” This mechanism then motivates the development of a simpler but more effective BQ method informed by the exponential impact. This BQ method is equipped with measures of prediction uncertainties and active learning, driven by two new acquisition functions, which have insightful interpretations, closed-form expressions and sound performance. Further, a transitional learning scheme based on adaptive tempering is developed and embedded into the developed BQ method, allowing for adaptive inference of sharp posteriors with desired accuracy. Several specific treatments, including elimination of error accumulation across stages, adaptive specification of tempering parameters, etc., have been developed for achieving the robust and efficient Transitional Bayesian Quadrature (TBQ) algorithm. Ultimately, the performance of TBQ for learning posteriors with multiple disconnected modes, high sharpness and nonlinear dependencies are demonstrated with both numerical cases and dynamics models.

1. Introduction

Numerical simulation has been widely accepted as a central tool in modern computational science for modeling multi-physical processes, as well as assisting decision-making and design of products in all engineering areas. Due to assumptions for simplification, discretization errors, unknownness of physical parameters, etc., those numerical simulators often struggle to accurately predict the responses of physical processes. This has motivated the proposition of the model Verification, Validation and Uncertainty Quantification (VV&UQ) framework for addressing the credibility of numerical simulators [1,2]. Inverse UQ, which aims at refining the simulators, or selecting the best simulator from a set of candidates, or averaging these candidates, to achieve a acceptable consistence between model predictions and observations of the physics, is one of the core theoretical subtasks of this framework. In this context, Bayesian model updating (BMU) has emerged as the most appealing gold-standard paradigm [3,4], due to its great potential for addressing the ubiquitous ill-posedness, and also its intrinsic ability of using probabilistic models to quantify various

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sources of uncertainties which result in the prediction disagreement. The later further allows decision-making and design with controllable risk tolerance. Beyond this, the BMU framework has also seen prospective potentials in many other areas such as inference of material parameters [5], monitoring of structural damages [6], source identification for dynamic physical fields [7], and endowing AI models with interpretability and uncertainty quantification [8]. The numerical computational challenge needs to be addressed is mainly on the estimation of the posteriors and the associated model evidence, given that the likelihoods are expensive to evaluate and the posteriors show complex behaviors.

The past few decades have witnessed prosperous developments of multiple branches of methods towards this challenge [9], which include, approximate analytical methods, Markov Chain Monte Carlo (MCMC) sampling methods, variational Bayesian inference methods, Bayesian model Updating with Structural reliability (BUS) methods, transport map, Bayesian filter, and Bayesian Quadrature (BQ). The approximate analytical methods, like Laplace's method and its many variants [10,11], are one class of classical methods for estimating model evidence. These methods employ a second-order Taylor expansion around the global maximum of the posterior, and thus can be effective for problems with sharp posteriors. While demonstrating good suitability to low-dimensional problems, these methods require integration with, e.g., variational inference or sampling methods for high-dimensional and/or strongly non-Gaussian problems [12].

MCMC sampling is perhaps the most renowned class of advanced approaches for BMU, which are based on approaching the target posterior density with the stationary distribution of one or multiple parallel Markov chains [13]. Typical methods in this group include Metropolis–Hastings (MH) sampling [14], Hamiltonian Monte Carlo sampling which performs proposal sampling by solving an equivalent Hamiltonian dynamics equation [15], transitional MCMC which aims at approaching the target posterior by performing MH sampling sequentially at a series of tempering stages [16], etc. Benefiting from the theoretical guarantee of convergence, the flexibility for improvements to address different scenarios, the suitability to high-dimensional cases, and the ease of parallelization, this group of methods have received the most attention. However, due to the extensive number of required model calls, these methods may face challenge when applied to expensive-to-evaluate simulators.

Variational Bayesian inference is based on postulating a parameterized probability distribution model, such as a Gaussian mixture model, for the target posterior, and then estimating the variational parameters by minimizing the Kullback–Leibler (KL) divergence between variational and target distributions, or equivalently, by maximizing the so-called evidence lower bound, this way to approach the target density [17,18]. These methods commonly require pre-specification of a parameterized distribution, for which a proper assumption is problem-dependent. They are commonly encouraged for problems with clearly defined model structures, and are less applicable to problems with unknown and complex posteriors.

The fundamental rationale behind BUS is to first transform the BMU problem to an equivalent structural reliability analysis problem, based on the principle of rejection sampling, and then solve it numerically with a well-established structural reliability method [19]. This framework has paved the way for several specific developments, especially those on sampling [20] and active learning [21]. For many BMU problems, the equivalent structural reliability problems may pose challenges due to the associated small failure probability and perhaps multiple disconnected failure domains [22].

Transport map has emerged in the past decade as an alternative approach, aiming to train an invertible map, equivalent to the Rosenblatt transformation, between a reference distribution and the posterior, allowing to transport an arbitrary number of samples from the reference to the posterior [23,24]. This group of methods require solving an optimization problem by minimizing the KL divergence between target and pullback densities, which can sometimes itself be numerically challenging.

Bayesian filter algorithms, such as unscented Kalman filter [25] and particle filter [26], have also been introduced for addressing BMU, which are either less applicability to multimodal posterior or challenged by particle degeneracy, though can be alleviated by specific skills.

A common feature of the above-mentioned methods is that, with specific treatments, they may be applicable for problems with complex posteriors, but commonly require huge number of model calls, which weakens their applicability to expensive-to-evaluate simulators. Among the past decade, some variants of BQ methods have also been developed, and show remarkable competitiveness in terms of efficiency, see e.g., Refs. [27–29]. These methods reformulate the integral equation, with which the model evidence is defined, with a probabilistic description, and then infer a posterior probabilistic description for predicting the model evidence. A Gaussian process (GP) model is trained and updated with sequentially designed training points to achieve desired accuracy. These BQ methods differ from each other in terms of the proxy functions being approximated by the GP model as well as the schemes for data acquisition, and most show good performance for capturing multimodality of posteriors. The aim of this work is to extend the BQ method for tackling complex posteriors with, e.g., multiple disconnected modes, sharply peaked modes and/or nonlinear dependencies. These tricky features exist in many multiphysical simulators with highly nonlinear/monotonic behaviors, deep gradients and high-quality observations. One can see Fig. 1 for an example. Estimation of posterior densities and the associated model evidences with high efficiency and desired error tolerance has consistently been a tough problem, even for the existing BQ methods.

To fill the above gap, improvements in multiple aspects are proposed in this work. First, the intrinsic mechanism behind the unsurprising performance of the BQ methods for multimodal posterior is revealed, from which it is concluded that the BQ method can further benefit from the exponential impact. Then, inspired by the revealed exponential impact, a new BQ method, equipped with two new acquisition functions for adaptive design of training points, and a new scheme for measuring prediction uncertainties, is developed for efficiently addressing the BMU problems subjected to multi-modal and nonlinearly dependent posteriors. Ultimately, this BQ method is combined with the TMCMC scheme for tackling BMU problems with also sharply peaked posteriors. Several specific treatments are developed, with the aim of avoiding error accumulation across consecutive tempering stages, acquiring higher transitional efficiency and achieving posterior samples of high quality. All these elaborate developments ultimately give rise

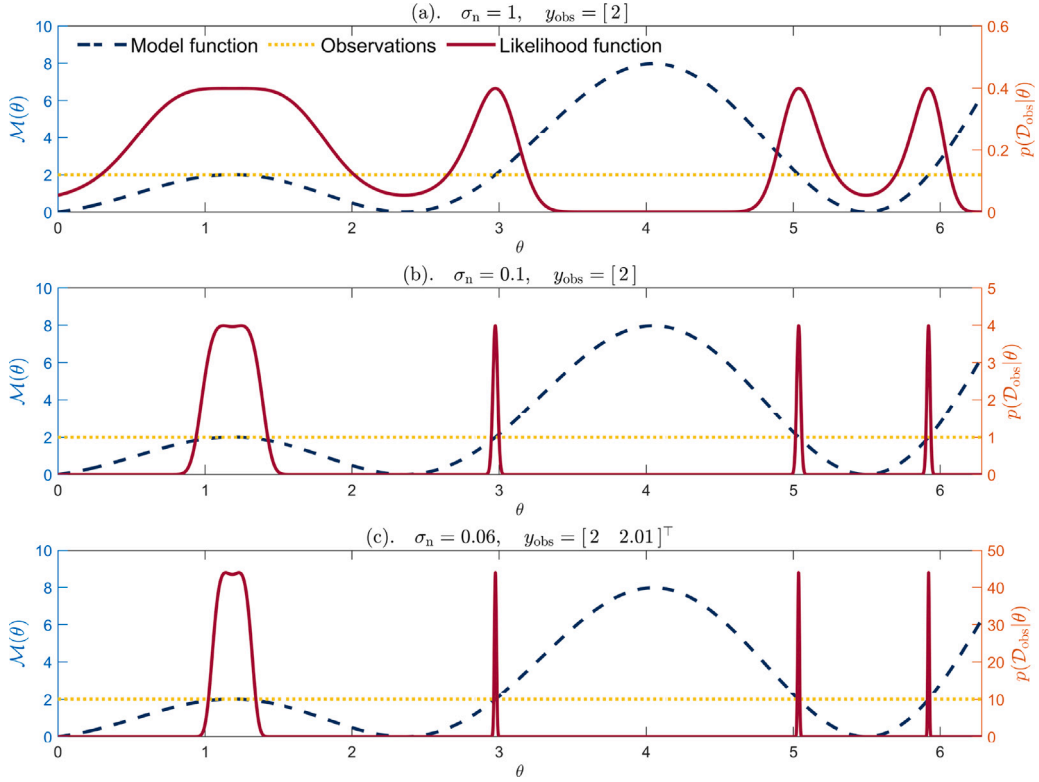


Fig. 1. Illustration of the challenge with multi-modal likelihood and high model evidence, where the left vertical axis indicates the values of model response, and the left one is for likelihood function. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

to a highly robust algorithm called Transitional Bayesian Quadrature (TBQ), which is shown to be of high efficiency for posteriors with multiple disconnected, sharply peaked and nonlinearly dependent modes.

The rest of this article is organized as follows. Section 2 presents the statement of BMU problem, and also explains the challenges concerning the multimodality and sharpness of the posteriors, followed by brief review of the BQ methods for BMU problem in Section 3. In Section 4, the intrinsic mechanism is first explained, followed by two new measures of prediction uncertainties, two new acquisition functions and the resultant BQ method. Section 5 presents the TBQ algorithm, including the adaptive specification of tempering parameters, the estimators for (intermediate) posteriors and evidences, the stopping conditions, the sampling scheme, and ultimately, the detailed description of the TBQ algorithm. Demonstration of the performance of the proposed methods with numerical and engineering examples is presented in Section 6. Finally, Section 7 summaries the crucial aspects generated from theoretical analysis and numerical demonstrations, and presents a prospect for future research.

2. Mathematical formulation of Bayesian model updating and challenges

Let $\mathcal{M}(\mathbf{x}, \theta)$ denote the simulation model, where $\mathbf{x} = (x_1, x_2, \dots, x_n)^\top$ refers to the n -dimensional controllable input vector consisting of e.g., spatial/temporal variables and external excitation, and $\theta = (\theta_1, \theta_2, \dots, \theta_d)^\top$ indicates the d -dimensional vector of model parameters, which is assumed to be deterministic but unknown, and needs to be calibrated based on observations. Denote by $\mathcal{D}_{\text{obs}} = \left\{ \left(\mathbf{x}_{\text{obs}}^{(j)}, y_{\text{obs}}^{(j)} \right) \right\}_{j=1}^{N_{\text{obs}}}$ a set of N_{obs} observations, where $\mathbf{x}_{\text{obs}}^{(j)}$ is the location where the j th output value $y_{\text{obs}}^{(j)}$ is observed. Further, denote by $\epsilon^{(j)}$ the measurement uncertainty at $y_{\text{obs}}^{(j)}$, which, following the common treatment, is assumed to be Gaussian white noise with zero mean and variance σ_n^2 , and thus is simply denoted as $\epsilon^{(j)} \equiv \epsilon \sim \mathcal{N}(0, \sigma_n^2)$. One notes the developed methods are not limited by the assumption of Gaussian white noise, and applies to noise with any type of distribution. Then, the relationship between observations and model predictions is formulated as:

$$y_{\text{obs}}^{(j)} = \mathcal{M}(\mathbf{x}_{\text{obs}}^{(j)}, \theta) + \epsilon, \quad (1)$$

with $j = 1, 2, \dots, N_{\text{obs}}$.

Based on the assumption of Gaussian white noise and independence of the measurements, the likelihood function of the measurements D_{obs} , conditioning on θ , are formulated as:

$$p(D_{\text{obs}}|\theta) = \prod_{j=1}^{N_{\text{obs}}} \phi(y_{\text{obs}}^{(j)} - \mathcal{M}(x_{\text{obs}}^{(j)}, \theta) | 0, \sigma_n^2), \quad (2)$$

where $\phi(\cdot | 0, \sigma_n^2)$ refers to the univariate Gaussian density with zero mean and constant variance σ_n^2 . In this work, the value of σ_n^2 is assumed to be exactly known for simplicity. In practical applications where this value is unknown, it can be treated as a model parameter, and assigned a prior distribution (e.g., inverse-Gamma distribution, log-normal distribution or the non-informative Jeffreys prior) [12,30], this way to include it in θ for inference.

Then, given a prior density $p(\theta)$ reflecting the analysts' prior knowledge or the physical meanings of the parameters θ , the posterior density of θ conditioning on D_{obs} is formulated, based on the Bayesian formula, as:

$$p(\theta|D_{\text{obs}}) = Z^{-1} p(D_{\text{obs}}|\theta) p(\theta), \quad (3)$$

where Z is a constant defined as the integral of the likelihood function over the prior density, i.e.,

$$Z = \int_{\mathbb{T}} p(D_{\text{obs}}|\theta) p(\theta) d\theta, \quad (4)$$

given $\mathbb{T} \subseteq \mathbb{R}^d$ is the support of the prior density. The constant Z is called “model evidence”, or “normalizing constant”, or “marginal likelihood”, as it reflects the credibility of the model, and can also used for model selection and model averaging [9]. The negative logarithm of the evidence Z , i.e., $-\log Z$ measures the statistical surprise, defined as the extent of unlikelihood of the observation D_{obs} under the model $\mathcal{M}$.

Due to the non-linearity of the simulation model $\mathcal{M}(x, \theta)$ over the model parameters, as well as the randomness of measurement noise, it is usually impossible to devise the analytical expressions of the posterior density $p(\theta|D_{\text{obs}})$, the model evidence Z , and the marginal posterior density of a subset model parameters, which are required in some application scenarios. This is mainly due to the difficulty of estimating the integral of Eq. (4) as each call of the integrand (likelihood) requires a call of the expensive-to-evaluate simulation model $\mathcal{M}(x, \theta)$. This numerical computational task faces a more severe challenge as the likelihood, and thus also the posterior, show multi-modal and sharply peaked behaviors. I use a simple example to illustrate this phenomenon, as shown in Fig. 1.

Assume an input-independent simulation model with only one model parameter, as shown by the dark-colored dashed line in Fig. 1. For Fig. 1(a), it is assumed that one observation is measured as $y_{\text{obs}}^{(1)} = 2$ with the noise Standard Deviation (STD) being $\sigma_n = 1$. The resultant likelihood is shown with the solid line. As can be seen, due to the non-monotonicity of the model response over θ , the likelihood function show multi-modal behavior, and due to the high noise variance, the likelihood show large variation around each mode, especially the one around $\theta = 1$. Comparing these modes it also found that larger gradients leads to sharper modes. With the same observation but smaller noise STD $\sigma_n = 0.1$, the likelihood function is shown in Fig. 1(b), which shows that, with smaller measurement error, the likelihood gets more sharply peaked. Further, with one more measurement $y_{\text{obs}}^{(2)} = 2.01$ added, and the noise STDs of both measurements assumed to be 0.06, each mode becomes even more sharply peaked, as indicated by Fig. 1(c). For the three scenarios shown in Fig. 1, the model evidences are sequentially 0.144, 0.271 and 1.8014, showing an increasing trend, indicating increasing credibility of models after calibrated. Estimation of multi-modal and sharp posterior density and the associated evidence, with desired accuracy, is a challenging task due to the substantial divergence between prior and posterior densities and the sparsely distributed modes.

3. Review of Bayesian quadrature for Bayesian model updating

MCMC sampling methods are the most commonly suggested schemes for solving the above BMU problem [13], but are also known to be less effective for problems with multi-modal and sharp posteriors, due to the low probability of transferring from one mode to another [13,16]. Although having been improved for such kind of tricky scenarios using e.g., adaptive tempering [16] or parallel tempering [31], these methods usually require huge number of likelihood calls, which can be unaffordable for problems with expensive-to-evaluate simulators. Alternatively, the BQ method has emerged as a promising strategy for tackling low- to medium-dimensional problems with multimodal posteriors, as it commonly requires much less model calls, to achieve a desired accuracy, than the sampling methods. However, there remains considerable scope for improvement in this method, in terms of both efficiency and suitability, especially for problems with multimodal, sharply peaked and nonlinear dependent posteriors.

In short, BQ formulates a quadrature rule using Bayesian inference scheme based on an arbitrary set of quadrature nodes, and presents a probabilistic estimate for the integral equation, this way to also quantify the prediction uncertainty. Take the following integral equation as an example:

$$I = \Pi[g(\theta)] = \int_{\mathbb{R}^d} g(\theta) p(\theta) d\theta, \quad (5)$$

where $g(\theta)$ is an expensive-to-evaluate integrand, and θ refers to the vector of integral variables. The standard BQ scheme is initialized by assuming a prior GP model $\hat{g}(\theta) \sim \mathcal{GP}(m(\theta), \kappa(\theta, \theta'))$ for the integrand, where $m(\cdot) : \mathbb{R}^d \rightarrow \mathbb{R}$ and $\kappa(\cdot, \cdot) : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ are the prior mean function and covariance function respectively, pre-specified by the users. This results in a prior Gaussian distribution $\hat{I} \sim \mathcal{N}(\mu_I, \sigma_I^2)$ for describing I , where $\mu_I = \Pi[m(\theta)]$ and $\sigma_I^2 = \Pi' \Pi [\kappa(\theta, \theta')]$, with $\Pi' \Pi[\cdot] = \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} p(\theta) p(\theta') d\theta d\theta'$.

Assume now a set of quadrature points $D_{N,\text{train}} = \{\mathcal{T}, \mathcal{Y}\} = \{(\theta^{(k)}, y^{(k)})\}_{k=1}^N$ are available, where $\mathcal{T}$ is a matrix of dimension $N \times d$ with its k th row being $\theta^{(k)}$, and $\mathcal{Y}$ a column vector with its k th element being $y^{(k)} = g(\theta^{(k)})$. One notes that those training points are commonly generated by an active learning scheme, meaning that they are dynamically generated, and then labeled with the desired responses, by interactively querying an acquisition function about the most valuable point. Thereof, the performance of the BQ scheme is primarily governed by the utilized acquisition function. To distinguish between the prior and posterior, I denote the posterior GP model as $\hat{g}_N(\theta) \sim \mathcal{GP}(\mu_{g,N}(\theta), c_{g,N}(\theta, \theta'))$, with the subscript “ N ” indicating that it is trained with N points. The posterior mean $\mu_{g,N}(\theta)$, variance $\sigma_{g,N}^2(\theta)$ and covariance $c_{g,N}(\theta, \theta')$ can be derived from the Bayesian rule:

$$p(\hat{g}_N(\theta) | \mathcal{Y}) \stackrel{\hat{g}_N = \hat{g}}{=} \frac{p(\mathcal{Y} | \hat{g}(\theta)) p(\hat{g}(\theta))}{p(\mathcal{Y})} = \frac{p(\mathcal{Y}, \hat{g}(\theta))}{p(\mathcal{Y})} \quad (6)$$

as [32]:

$$\mu_{g,N}(\theta) = m(\theta) + \kappa(\theta, \mathcal{T}) \mathcal{K}^{-1} (\mathcal{Y} - m(\mathcal{T})) \quad (7a)$$

$$\sigma_{g,N}^2(\theta) = \kappa(\theta, \theta) - \kappa(\theta, \mathcal{T}) \mathcal{K}^{-1} \kappa(\mathcal{T}, \theta) \quad (7b)$$

$$c_{g,N}(\theta, \theta') = \kappa(\theta, \theta') - \kappa(\theta, \mathcal{T}) \mathcal{K}^{-1} \kappa(\mathcal{T}, \theta'), \quad (7c)$$

where $\kappa(\theta, \mathcal{T})$ is a row vector with its k th element being $\kappa(\theta, \theta^{(k)})$, $\kappa(\mathcal{T}, \theta)$ is the transposition of $\kappa(\theta, \mathcal{T})$, $m(\mathcal{T})$ is a column vector with the k th element being $m(\theta^{(k)})$, and $\mathcal{K}$ is the Gram matrix of dimension $N \times N$, with its (k, l) th element being $\kappa(\theta^{(k)}, \theta^{(l)})$. Replacing the integrand g in Eq. (5) with $\hat{g}_N$ yields a posterior Gaussian variable $\hat{I}_N \sim \mathcal{N}(\mu_{I,N}, \sigma_{I,N}^2)$ for summarizing the prediction of I , with the posterior mean $\mu_{I,N}$ and variance $\sigma_{I,N}^2$ expressed by [33,34]:

$$\mu_{I,N} = \mu_I + \Pi [\kappa(\theta, \mathcal{T})] \mathcal{K}^{-1} (\mathcal{Y} - m(\mathcal{T})) \quad (8a)$$

$$\sigma_{I,N}^2 = \sigma_I^2 - \Pi [\kappa(\theta, \mathcal{T})] \mathcal{K}^{-1} \Pi' [\kappa(\mathcal{T}, \theta')] \quad (8b)$$

Mathematically, the posterior mean $\mu_{I,N}$ presents a mean prediction of I , and the posterior variance $\sigma_{I,N}^2$ summarizes the prediction uncertainty. The second term $\Pi [\kappa(\theta, \mathcal{T})] \mathcal{K}^{-1} \Pi' [\kappa(\mathcal{T}, \theta')]$ in Eq. (8b) measures the amount of reduction of the prediction uncertainty, compared to the prior.

As revealed by Eq. (4), the model evidence Z is also formulated as a d -dimensional integral with the integrand being the likelihood function. This may motivate the utilization of the above standard BQ method for estimating the model evidence, which will also result in an estimate of the posterior density with desired accuracy. However, this has been repeatedly shown to be impractical, or at least, inefficient (see e.g., Refs. [27,28,35,36]). This is mainly due to the multi-modal, sparse and sharp features of the likelihood function (see Fig. 1), making it difficult to be captured by a GP model based on relatively small number of training points, and thus also ineffective for active inference. This has motivated the developments of alternative approximate BQ schemes to fill this gap. Some typical developments are summarized in Table 1. Most of these developments, such as Refs. [27,28,35], are developed and published in the area of machine learning, and lacks sufficient visibility in the area of uncertainty quantification for simulation of physical processes and fields. As it is impractical to approximate the likelihood function with a GP, these developments built different functions connected to the likelihood, called proxy functions, being fitted by the GP models, and these include logarithm of likelihood [18,35], square root of likelihood [27,28], and model function $\mathcal{M}$ [36]. For the former two kinds of proxy functions, approximation schemes, such as linearization of the likelihood over the proxy function, and moment matching, have been developed, allowing for describing the answers to the model evidences as Gaussian variables with closed-form expressions, and also deriving closed-form expressions of the acquisition functions for efficient active inference of training points. Considering that the linearization and moment-matching schemes may cause uncontrollable risk of accuracy loss, the sampling-based inference scheme based on random functional samples of the GP model, has been developed, but the sampling scheme can be less efficient in terms of computer time, and introduces extra approximate error [29,36]. Alternatively, the variational Bayesian inference scheme is utilized in Ref. [18] to transform the estimation of posterior density as an optimize-then-integrate nested problem, and then Bayesian optimization, with Expected Improvement (EI) function as acquisition function, and the exact Bayesian quadrature, with the Posterior Variance Contribution (PVC) function as acquisition function, are combined in a sequential way for inferring the posterior density with desired accuracy. In this work, three contributions towards improving BQ for problems with tricky posteriors are presented. First, a mechanistic explanation is presented to address why the BQ method is effective and how it can be further improved to enhance its efficacy. Second, inspired by the observed mechanisms, a new BQ algorithm method, equipped with new scheme for quantifying the prediction uncertainties, and two new acquisition functions for more effective design of training points, is developed. Third, the TBQ method, which combines TCMC and BQ, is developed for addressing BMU problems with multimodal, sharply peaked and nonlinear dependent posteriors with high efficiency, accuracy and robustness.

4. A new Bayesian quadrature method for multimodal posteriors

4.1. Exponential impact and intrinsic mechanism

The first proposed method, called exponential impact boosted Bayesian Quadrature, abbreviated as BQ for simplicity, is inspired by taking the logarithm of the likelihood as the proxy function to be approximated by the GP model, i.e., $g(\theta) = \log p(D_{\text{obs}} | \theta)$. The

Table 1

Summary of selected approximate BQ methods for BMU.

Sources	Proxy functions fitted by GP	Schemes for inference	Acquisition functions
Ref. [35]	$\log p(D_{\text{obs}} \theta)$	Linearization	US ^b
Ref. [27]	$\sqrt{2(p(D_{\text{obs}} \theta) - \alpha)}$ ^a	Linearization or moment matching	US
Ref. [28]	$\sqrt{2(p(D_{\text{obs}} \theta) - \alpha)}$	Linearization	Importance US
Ref. [36]	$\mathcal{M}(x, \theta)$	Sampling of GP	PVC ^c
Ref. [18]	$\log p(D_{\text{obs}} \theta)$	Variational inference & Bayesian optimization	EI ^d and PVC
Ref. [29]	$\log p(D_{\text{obs}} \theta)$	Sampling of GP	PVC

^a α is a small positive constant suggested to be $0.8 \times \min p(D_{\text{obs}}|\theta)$.^b US refers to the Uncertainty Sampling function, derived from the posterior variance of the GP model.^c PVC is the Posterior Variance Contribution function, devised with the posterior covariance of the GP model [34].^d EI implies the Expected Improvement acquisition function for Bayesian optimization [37].

readers can skip this subsection if only the methodology details are of interest; while if, an in-depth understanding on the motivation of the developments and other closely related concepts are of concern, discussions in this subsection can be helpful.

It has been repeatedly shown that approximating $\log p(D_{\text{obs}}|\theta)$ with a GP can provide better performance to the likelihood than approximating the likelihood $p(D_{\text{obs}}|\theta)$ itself, see e.g., Ref. [35]. Herein, the likelihood shown in Fig. 1(b) is utilized for illustrating this experience, as the likelihood in this case is much more sharply peaked than that in Fig. 1(a), but with its logarithm easier to be approximated by a GP model than Fig. 1(c). Twenty training samples of θ are generated by Latin-hypercube sampling (LHS). With the values of likelihood function as labels, the posterior mean predication and the associated credible interval (CI) of the resultant GP model are shown in Fig. 2(a), together with the exact likelihood function for comparison. Besides, with the values of the proxy function $g(\theta)$ as labels, the mean prediction and the 95% CI of the resultant GP model $\hat{g}_N(\theta)$ are shown in Fig. 2(b), together with the true curve of $g(\theta)$; and then the resultant probabilistic surrogate model of the likelihood, generated by the inverse transformation $p(D_{\text{obs}}|\theta) = \exp(g(\theta))$, is presented in Fig. 2(c). One notes that the CI shown in Fig. 2(b) is so narrow that it almost overlaps with the mean prediction. It is clearly seen by Fig. 2(c) that, all the four modes have been properly captured by a GP model $\hat{g}_N(\theta)$, and the prediction uncertainty for all modes are also properly described. Whereas, the GP model shown in Fig. 2(a) fails to capture the three sharp modes. For the leftmost flat mode, although it is captured, it shows relative higher bias, and the prediction uncertainty is underestimated. The proper capture of all the four modes by $\hat{g}_N(\theta)$ is mainly benefited from the innocuous behavior of the proxy function, even given that the likelihood show multimodal and sharply peaked behaviors.

Why it works? One can build an intuition behind this by comparing the exact curves of the likelihood and the proxy function shown in Fig. 2. For the likelihood function shown in Fig. 2(a), the modes are sparsely distributed in the prior support $[0, 2\pi]$, meaning that, in most region, it admits zero value. The behavior in modes' supports definitely cannot be revealed by the training points in the empty region, even in case these training points are very close to the four modes' supports. However, this curse can be substantially alleviated by approximating the logarithm of likelihood. It is seen in Fig. 2(c) that, the mode around $\theta = 3$ is approximated with low bias but high variation, even there is no training point located within the support of this mode. This benefit stems from the considerably less sharpness of the proxy function, which enables the revelation of features within the supports of the four modes, even with training points spatially outside these supports. The above mechanism can also be further validated by the length-scale parameters of the kernel function used in training the GP model, the values of which reflect the magnitude of length-scale correlations [32,38]. Specifically, the two GP models shown in Fig. 2(a) and (b) are both trained with squared exponential kernel, and the values of the length-scale parameters are computed by maximum likelihood estimation to be 0.2174 and 0.5478 respectively (see chapter 5 of Ref. [32] for details). Indeed, in case the length-scale approaches zero, the kernel degrades into the Dirac kernel, and the BQ rule collapses to the crude Monte Carlo (MC) quadrature rule, whose convergence rate is $\mathcal{O}(N^{-1/2})$. This means that it requires a vast number of samples to reach convergence.

A comparison of Fig. 2(b) and (c) reveals that, given the small prediction uncertainty of the proxy function, it is amplified in the supports of the four modes when mapping it to approximate the likelihood. This arises from the multiplier effect of the exponential transformation, which paves the way for further development. Specifically, this facilitates more effective active learning, whereby new training points are more likely to be acquired in such regions to achieve maximum reduction in prediction uncertainties for both posteriors and model evidences. Beyond this, the exponential multiplier effect will shrink the prediction uncertainty and bias in the empty region of likelihood while mapping the GP model from the proxy function to the likelihood, as revealed by the rightmost region of Fig. 2(b) and (c). This advantage specifically prevents the algorithm from generating training points in those uninformative regions. A further advantage of this treatment is that it inherently guarantees non-negative lower bounds for both likelihood and posterior, since the exponential of any real number is non-negative. Thus, the lower bound shown in Fig. 2(c) is truncated at zero. In summary, using the logarithm of likelihood as a proxy function facilitates more effective learning of the likelihood and offers potential advantages for optimizing the design of training points.

It is also noted that some of the developments shown in Table 1, while also approximating the logarithm of likelihood with GP, do not fully exploit the exponential multiplier effect. For example, by linearization of this exponential transformation, such as the one treated in Ref. [35], the above-mentioned advantages will be weakened. In my previous work [18], the logarithm of likelihood was taken as the proxy function, and then with the variational Bayesian inference scheme, the problem is formulated as an optimize-then-integrate nested problem with the integrand being the proxy function. This further enables efficient and exact Bayesian inference,

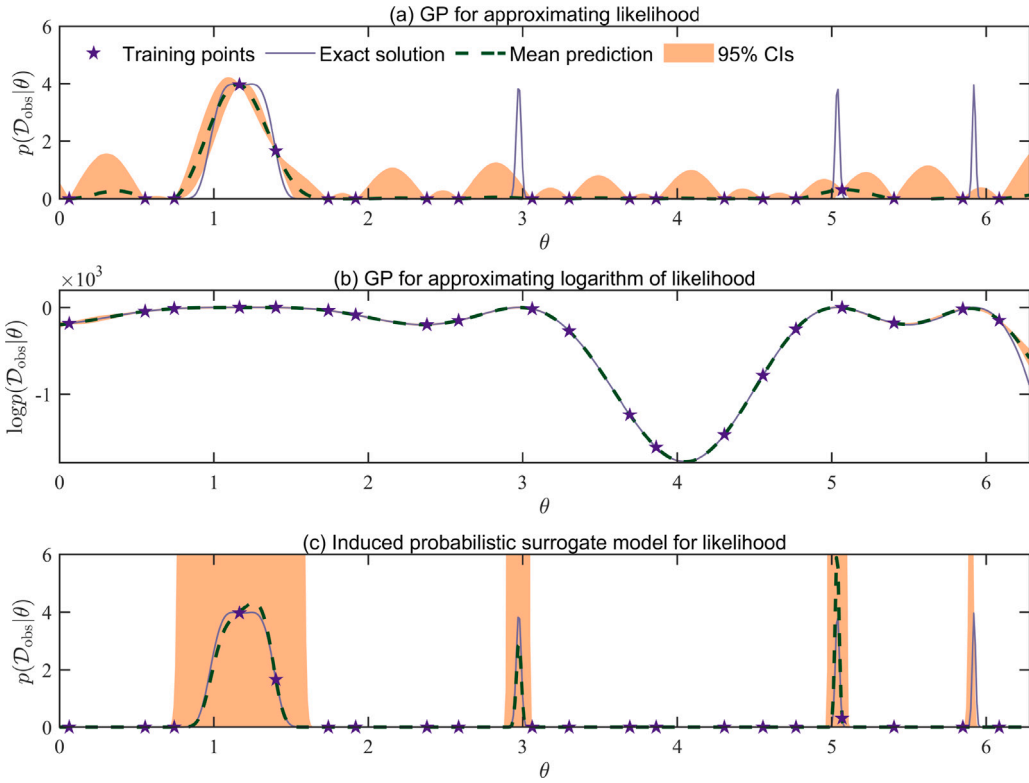


Fig. 2. Comparison of two alternative schemes for approximating likelihood with GP model, where (a) shows the GP model trained for the likelihood, (b) presents the GP model trained for the proxy function, and (c) gives the probabilistic surrogate model generated by the exponential transformation of the GP presented in (b).

as closed-form expressions of acquisition functions in both Bayesian optimization and Bayesian quadrature are available. However, it was found empirically from benchmark studies that, the method, although is much more efficient than MCMC, requires higher number of model calls than expected. Through the mechanism explained above, the rationale behind this phenomenon is evident: when using the logarithm of likelihood as the integrand rather than its exponential transformation, the exponential effect remains unexploited in active learning.

The above mechanistic interpretations provide a basis for developing more effective BQ algorithms from various aspects, including prediction uncertainty measure, the design of more effective acquisition functions, and the potential for learning posteriors with complex behaviors.

4.2. Quantification of prediction uncertainties

Given the proxy function $g(\theta)$ being approximated by a GP model $\hat{g}_N(\theta) \sim \mathcal{GP}(\mu_{g,N}(\theta), c_{g,N}(\theta, \theta'))$, a lognormal process model can be devised using the inverse transformation $\hat{p}_N(D_{\text{obs}}|\theta) = \exp(\hat{g}_N(\theta))$. The linearization of this inverse transformation is utilized in Refs. [35], and a sampling scheme is utilized in Refs. [29,36], to approximately derive the posterior features of $\hat{p}_N(D_{\text{obs}}|\theta)$. I present an alternative scheme, which is inspired by the low confidence bound for Bayesian optimization active learning [39], and also the integrated margin probability function, originally developed for Bayesian active learning of failure probability [40].

Due to the Gaussianity of $\hat{g}_N(\theta)$, its CI prediction with credible level $100(1 - \alpha)\%$ is formulated as:

$$[\underline{g}_{N,\alpha}(\theta), \bar{g}_{N,\alpha}(\theta)] \triangleq [\mu_{g,N}(\theta) - \beta\sigma_{g,N}(\theta), \mu_{g,N}(\theta) + \beta\sigma_{g,N}(\theta)], \quad (9)$$

where $\beta = \Phi^{-1}(1 - \alpha/2)$, with $\Phi(\cdot)$ being the cumulative distribution function (CDF) of standard Gaussian distribution. Benefiting from the monotonicity of the exponential transformation, an equivalent CI can be derived for $\hat{p}_N(D_{\text{obs}}|\theta)$ as:

$$[\underline{p}_{N,\alpha}(D_{\text{obs}}|\theta), \bar{p}_{N,\alpha}(D_{\text{obs}}|\theta)] \triangleq [\exp(\underline{g}_{N,\alpha}(\theta)), \exp(\bar{g}_{N,\alpha}(\theta))], \quad (10)$$

and then an CI for the unnormalized posterior density as $[\underline{p}_{N,\alpha}(D_{\text{obs}}|\theta), \bar{p}_{N,\alpha}(D_{\text{obs}}|\theta)] \times p(\theta)$. One notes that the CI formulated by Eq. (10) is not a valid CI, as both bounds may not be realization of the resultant stochastic process $\exp(\hat{g}_N)$, the posterior correlation information has been omitted. However, it can serve as conservative measures of prediction uncertainties, since, intuitively, given

θ fixed at any point, the probability that the true value of the likelihood at this point belongs to this CI should not be lower than $1 - \alpha/2$. Henceforth, this distinction will no longer be elaborated, and the resulting bounds will be simply referred to as CIs. The width of the interval given by Eq. (10) can serve as a measure of prediction uncertainty associated with the likelihood, and is formulated by:

$$\mathcal{V}_{N,\alpha}^I(\theta) = \bar{p}_{N,\alpha}(D_{\text{obs}}|\theta) - \underline{p}_{N,\alpha}(D_{\text{obs}}|\theta). \quad (11)$$

It is called Prediction Uncertainty Quantification (PUQ) of type I. Similarly, an equivalent CI can be derived for the model evidence as $[\underline{Z}_{N,\alpha}, \bar{Z}_{N,\alpha}]$, with

$$\bar{Z}_{N,\alpha} = \int_{\mathbb{T}} \exp(\bar{g}_{N,\alpha}(\theta)) p(\theta) d\theta \quad (12a)$$

$$\underline{Z}_{N,\alpha} = \int_{\mathbb{T}} \exp(\underline{g}_{N,\alpha}(\theta)) p(\theta) d\theta. \quad (12b)$$

A mean prediction of the model evidence, denoted as $\mu_{Z,N}$, and bounded by $[\underline{Z}_{N,\alpha}, \bar{Z}_{N,\alpha}]$, is given by:

$$\mu_{Z,N} = \int_{\mathbb{T}} \exp(\mu_{g,N}(\theta)) p(\theta) d\theta. \quad (13)$$

One notes that $\mu_{Z,N}$ is not the exact expectation of $\hat{Z}_N = \int_{\mathbb{T}} \exp(\hat{g}_N(\theta)) p(\theta) d\theta$, which should be $\int_{\mathbb{T}} \exp(\mu_{g,N}(\theta) + \sigma_{g,N}^2(\theta)/2) p(\theta) d\theta$. However, in case of negligible prediction uncertainty, these two show small difference. One can use either one for mean prediction. For simplicity, Eq. (13) is utilized in what follows, as for methods developed in this work, the mean prediction is required only after the algorithm is converged with pre-specified prediction accuracy.

An alternative measure of the prediction uncertainty, called PUQ of type II, is defined by the quotient of the upper and lower bounds. For likelihood and posterior density, informed by Eq. (10), the PUQ of type II is defined as:

$$\mathcal{V}_{N,\alpha}^{\text{II}}(\theta) = \frac{\exp(\bar{g}_{N,\alpha}(\theta))}{\exp(\underline{g}_{N,\alpha}(\theta))} = \exp(2\beta\sigma_{g,N}(\theta)), \quad (14)$$

which is an exponential transformation of the posterior STD $\sigma_{g,N}$, and is independent of the mean $\mu_{g,N}$. Given the same credible level, the measures in Eqs. (11) and (14) quantify the amount of prediction uncertainty associated with the likelihood, and the integral of each over the prior density measures the prediction uncertainty associated with the model evidence.

It is seen that, given a GP model readily trained, the CI $[\underline{p}_{N,\alpha}(D_{\text{obs}}|\theta), \bar{p}_{N,\alpha}(D_{\text{obs}}|\theta)]$ of the likelihood, as well as the two uncertainty measures $\mathcal{V}_{N,\alpha}^I(\theta)$ and $\mathcal{V}_{N,\alpha}^{\text{II}}(\theta)$, all admit closed-form expressions. However, neither the CI in Eq. (12) nor the mean prediction in Eq. (13) has an analytical formulation, and their evaluation must rely on numerical algorithms. MC integration is suggested for this purpose. Assume that a set of N_{MC} random or quasi-random samples, denoted as $\mathcal{T}_{\text{MC}} = \{\theta_{\text{MC}}^{(k)}\}_{k=1}^{N_{\text{MC}}}$, is generated by, e.g., LHS or Sobol' low-discrepancy sequence [41], following the prior density $p(\theta)$, the MC estimators for the confidence bounds $[\underline{Z}_{N,\alpha}, \bar{Z}_{N,\alpha}]$ are formulated by:

$$\bar{Z}_{N,\alpha} = \frac{1}{N_{\text{MC}}} \sum_{k=1}^{N_{\text{MC}}} \exp(\bar{g}_{N,\alpha}(\theta_{\text{MC}}^{(k)})) \quad (15a)$$

$$\underline{Z}_{N,\alpha} = \frac{1}{N_{\text{MC}}} \sum_{k=1}^{N_{\text{MC}}} \exp(\underline{g}_{N,\alpha}(\theta_{\text{MC}}^{(k)})). \quad (15b)$$

Similarly, the MC estimator of the mean prediction $\mu_{Z,N}$ is given by:

$$\mu_{Z,N} = \frac{1}{N_{\text{MC}}} \sum_{k=1}^{N_{\text{MC}}} \exp(\mu_{g,N}(\theta_{\text{MC}}^{(k)})). \quad (16)$$

The variation of the above three MC estimators can be easily derived, and I do not give details.

A conservative CI for the posterior density $p(\theta|D_{\text{obs}})$ can then be estimated as:

$$[\underline{p}_{N,\alpha}(\theta|D_{\text{obs}}), \bar{p}_{N,\alpha}(\theta|D_{\text{obs}})] = [\bar{Z}_{N,\alpha}^{-1} \underline{p}_{N,\alpha}(D_{\text{obs}}|\theta), \underline{Z}_{N,\alpha}^{-1} \bar{p}_{N,\alpha}(D_{\text{obs}}|\theta)] p(\theta). \quad (17)$$

The mean prediction of the posterior density is then evaluated by:

$$p_N(\theta|D_{\text{obs}}) = \mu_{Z,N}^{-1} \exp(\mu_{g,N}(\theta)) p(\theta). \quad (18)$$

One notes that, the bounds estimated by Eq. (17) are neither valid densities, but the mean prediction computed by Eq. (18) is. The conservative CI computed by Eq. (17) is used in case studies to illustrate the accuracy of results.

4.3. Acquisition functions

Based on the two types of prediction uncertainty measures, two acquisition functions are developed in this subsection for adaptive design of the training points $\mathcal{D}_{N,\text{train}}$. The first one, inspired by the PUQ of type I, is defined as:

$$\begin{aligned}\mathcal{A}_{\text{PUQ},N,\alpha}^{\text{I}}(\theta) &= \mathcal{V}_{N,\alpha}^{\text{I}}(\theta) p(\theta) \\ &= \exp(\mu_{g,N}(\theta)) [\exp(\beta\sigma_{g,N}(\theta)) - \exp(-\beta\sigma_{g,N}(\theta))] p(\theta).\end{aligned}\quad (19)$$

It is seen that the integral of $\mathcal{A}_{\text{PUQ},N,\alpha}^{\text{I}}(\theta)$ over the support of prior is exactly the measure of prediction uncertainty of model evidence. Further, due to the nonnegativity of $\mathcal{A}_{\text{PUQ},N,\alpha}^{\text{I}}(\theta)$, it can also be interpreted as the contribution to prediction uncertainty. It can be observed that in the definition of $\mathcal{A}_{\text{PUQ},N,\alpha}^{\text{I}}(\theta)$, the exponential impact is explicitly incorporated for both posterior mean $\mu_{g,N}(\theta)$ and posterior STD $\sigma_{g,N}(\theta)$, enabling a trade-off between exploitation and exploration. Thus, the new training point is selected as $\theta^+ = \arg \max_{\theta \in \mathbb{T}} \mathcal{A}_{\text{PUQ},N,\alpha}^{\text{I}}(\theta)$.

Using the type I PUQ, the second acquisition function is formulated as:

$$\begin{aligned}\mathcal{A}_{\text{PUQ},N,\alpha}^{\text{II}}(\theta) &= \exp(\mu_{g,N}(\theta)) \mathcal{V}_{N,\alpha}^{\text{II}}(\theta) p(\theta) \\ &= \exp(\mu_{g,N}(\theta)) \exp(2\beta\sigma_{g,N}(\theta)) p(\theta),\end{aligned}\quad (20)$$

where the term $\exp(\mu_{g,N}(\theta))$ is presented for introducing exponential impact for the posterior mean $\mu_{g,N}(\theta)$, as the uncertainty measure given by Eq. (14) does not explicitly include $\mu_{g,N}(\theta)$. This enables the algorithm to make a balance between exploration and exploitation. The most valuable design point θ^+ can then be specified by maximizing $\mathcal{A}_{\text{PUQ},N,\alpha}^{\text{II}}(\theta)$. As will be shown in the test examples, the two acquisition functions show similar performance. However, theoretically, given high prediction uncertainty of the GP model, as $\exp(-\beta\sigma_{g,N}(\theta))$ approaches zero, the second acquisition function places a slightly greater emphasis on exploration compared to the first one.

Owing to the closed-form expressions of these two acquisition functions, their global maxima can be readily estimated by standard global optimization algorithms such as Particle Swarm Optimization (PSO) and Genetic Algorithm (GA), enabling numerically efficient adaptive design.

4.4. The developed Bayesian quadrature algorithm

Building on the above developments, an exponential-impact-informed BQ algorithm is presented in Algorithm 1. The inputs to this BQ algorithm, as shown in Algorithm 1, include the likelihood $p(\mathcal{D}_{\text{obs}}|\theta)$, the prior $p(\theta)$, the stopping threshold ϵ_α , the initial training sample size N_0 , the MC sample size N_{MC} and the number N_S for delayed stopping judgment. The outputs include the CI $[\underline{Z}_{N,\alpha,\text{MC}}, \bar{Z}_{N,\alpha,\text{MC}}]$ of the model evidence, the conservative CI $[\underline{p}_{N,\alpha,\text{MC}}(\theta|\mathcal{D}_{\text{obs}}), \bar{p}_{N,\alpha,\text{MC}}(\theta|\mathcal{D}_{\text{obs}})]$, the mean estimates $\mu_{Z,N}$ and $p_N(\theta|\mathcal{D}_{\text{obs}})$, as well as the total number N of the likelihood calls for illustrating the numerical efficiency of the algorithm.

The value of α , and thus β , plays a role for balancing the exploration and exploitation. With higher value of β , points with higher prediction uncertainty will be preferable, and thus resulting in preference to exploration performance; while with smaller value, the algorithm tend to exploit the local behavior of identified modes. Empirical experiences show that the performance of the algorithm is not very sensitive to the value of β , and thus it can be specified within a relative large range. Besides, it is suggested to set α as 0.05~0.6, and thus β as 0.5244~1.9600. The initial training sample size N_0 can be empirically set to be a value between 10 and 20, and N_{MC} is suggested to be $10^4 \sim 10^5$.

The stopping condition is informed by the relative error of model evidence as:

$$\frac{\bar{Z}_{N,\alpha} - \underline{Z}_{N,\alpha}}{\bar{Z}_{N,\alpha} + \underline{Z}_{N,\alpha}} < \epsilon_\alpha, \quad (21)$$

where ϵ_α is an α -dependent threshold, with smaller α (e.g., higher confidence), ϵ_α should be attributed a higher value. Generally, it is suggested to set 0.05~0.2 for Algorithm 1. Besides, to avoid fake convergence, which may appear in the early training stage, it is suggested use a delayed judgment scheme, which means to stop the algorithm when the stopping condition is satisfied N_S (1 or 2) times consecutively, as detailed in Algorithm 1.

5. Transitional Bayesian quadrature for sharply peaked posteriors

5.1. Basic idea of transitional inference

Empirical results demonstrate that Algorithm 1 is effective for moderately sharp multimodal posteriors, like the one illustrated in Fig. 1(a). For extremely sharp posteriors, such as the one shown by Fig. 1(b) and (c), this algorithm is less effective. This phenomenon becomes increasingly pronounced with increasing dimensionality. This stems from the limitations of the GP model trained with a small number of points in capturing sharp modes and estimating the associated model evidence while properly quantifying potential prediction uncertainties. The transitional learning is a critical inherent feature of Bayesian inference that distinguishes it from other statistical inference methods, and has been widely applied in many theoretical researches and practices, see, Refs. [16,29,31,42,43] for example. A practical transitional learning scheme involves introducing a tempering parameter (also known as a scale parameter)

Algorithm 1: The BQ algorithm

Input: $p(\mathcal{D}_{\text{obs}}|\theta)$, $p(\theta)$, α , ϵ_α , N_0 , N_{MC} , N_{S}

Output: $\left[\underline{Z}_{N,\alpha}, \overline{Z}_{N,\alpha}\right]$, $\left[\underline{p}_{N,\alpha}(\theta|\mathcal{D}_{\text{obs}}), \overline{p}_{N,\alpha}(\theta|\mathcal{D}_{\text{obs}})\right]$, $\mu_{Z,N}$, $p_N(\theta|\mathcal{D}_{\text{obs}})$, N

- 1 Initialization: StopFlag = 0, $N = N_0$;
- 2 $\mathcal{T}_{\text{MC}} = \left\{\theta_{\text{MC}}^{(i)}\right\}_{i=1}^{N_{\text{MC}}} \leftarrow$ LSH sampling from $p(\theta)$;
- 3 $\mathcal{T} = \left\{\theta^{(k)}\right\}_{k=1}^N \leftarrow$ LSH sampling from $p(\theta)$;
- 4 **while** StopFlag < N_{S} **do**
- 5 **if** $N = N_0$ **then**
- 6 $\mathcal{Y} \leftarrow \log p(\mathcal{D}_{\text{obs}}|\mathcal{T})$;
- 7 **else**
- 8 $\theta^+ \leftarrow \arg \max_{\theta \in \mathbb{T}} \mathcal{A}_{\text{PUQ},N,\alpha}(\theta)$;
- 9 $y^+ \leftarrow g(\theta^+) = \log p(\mathcal{D}_{\text{obs}}|\theta^+)$;
- 10 $\mathcal{T} \leftarrow \mathcal{T} \cup \{\theta^+\}$;
- 11 $\mathcal{Y} \leftarrow \mathcal{Y} \cup \{y^+\}$;
- 12 $N = N + 1$;
- 13 **end**
- 14 $\hat{g}_N(\theta) \leftarrow$ Train a GP with $\mathcal{D}_{N,\text{train}} = \{\mathcal{T}, \mathcal{Y}\}$;
- 15 $\left[\underline{Z}_{N,\alpha}, \overline{Z}_{N,\alpha}\right] \leftarrow$ Eq. (15);
- 16 $\mu_{Z,N} \leftarrow$ Eq. (16);
- 17 $\left[\underline{p}_{N,\alpha}(\mathcal{D}_{\text{obs}}|\theta), \overline{p}_{N,\alpha}(\mathcal{D}_{\text{obs}}|\theta)\right] \leftarrow$ Eq. (10);
- 18 $\left[\underline{p}_{N,\alpha}(\theta|\mathcal{D}_{\text{obs}}), \overline{p}_{N,\alpha}(\theta|\mathcal{D}_{\text{obs}})\right] \leftarrow$ Eq. (17);
- 19 $p_N(\theta|\mathcal{D}_{\text{obs}}) \leftarrow$ Eq. (18);
- 20 **if** Eq. (21) satisfied **then**
- 21 StopFlag = StopFlag + 1
- 22 **else**
- 23 StopFlag = 0;
- 24 **end**
- 25 **end**

into the likelihood, enabling a gradual transition from the prior to the posterior [16,31]. This scheme is combined with Algorithm 1 for transitional learning of complex posteriors in this section.

Specifically, given a tempering parameter $\gamma_j \in [0, 1]$, the corresponding intermediate posterior density is formulated as [16,31]:

$$p^{(j)}(\theta|\mathcal{D}_{\text{obs}}) = Z_j^{-1} [p(\mathcal{D}_{\text{obs}}|\theta)]^{\gamma_j} p(\theta), \quad (22)$$

where

$$Z_j = \int_{\mathbb{T}} [p(\mathcal{D}_{\text{obs}}|\theta)]^{\gamma_j} p(\theta) d\theta \quad (23)$$

is the intermediate model evidence associated with γ_j . It is straightforward to conclude that, with γ_j approaching zero, Z_j tends to assume the value of 1, meanwhile $p^{(j)}(\theta|\mathcal{D}_{\text{obs}})$ moves towards the prior $p(\theta)$; while with γ_j approaching one, Z_j gravitates toward the target model evidence Z , and $p^{(j)}(\theta|\mathcal{D}_{\text{obs}})$ moves toward the target posterior $p(\theta|\mathcal{D}_{\text{obs}})$. One can readily derive from Eqs. (22) and (23) that:

$$\frac{Z_{j+1}}{Z_j} = \int_{\mathbb{T}} [p(\mathcal{D}_{\text{obs}}|\theta)]^{\gamma_{j+1}-\gamma_j} p^{(j)}(\theta|\mathcal{D}_{\text{obs}}) d\theta, \quad (24)$$

indicating that the posterior defined by γ_j can be seen as a prior, and the increment $[p(\mathcal{D}_{\text{obs}}|\theta)]^{\gamma_{j+1}-\gamma_j}$ as a likelihood, to formulate the posterior $p^{(j+1)}(\theta|\mathcal{D}_{\text{obs}})$ associated with γ_{j+1} . This motivates the design of a sequence of values of γ , satisfying $0 = \gamma_1 < \gamma_2 < \dots < \gamma_M = 1$, to sequentially approach the target posterior with $p^{(j)}(\theta|\mathcal{D}_{\text{obs}})$ and the target model evidence with Z_j . With the likelihood shown in Fig. 1(b) as an example, by introducing four tempering stages, the resultant intermediate posterior densities are shown in Fig. 3. It is observed that the support of the posterior with lower γ_j covers that with higher γ_j , and as γ_j increases, the intermediate posterior contracts and exactly equals the target posterior when $\gamma_j = 1$. This is analogous to tempering, where in the initial high-temperature regime, particles become uniformly distributed due to increased kinetic energy, and as the temperature decreases, they converge to stable posterior modes.

Combining the BQ algorithm with this transitional learning scheme requires five critical developments. First, develop a strategy for **adaptive specification of the tempering parameter** γ_{j+1} . Second, develop **estimators** for both Z_{j+1} and $p^{(j+1)}(\theta|\mathcal{D}_{\text{obs}})$ (or its unnormalized version), as well as the equivalent CIs for quantifying the associated prediction uncertainties. Third, extend the

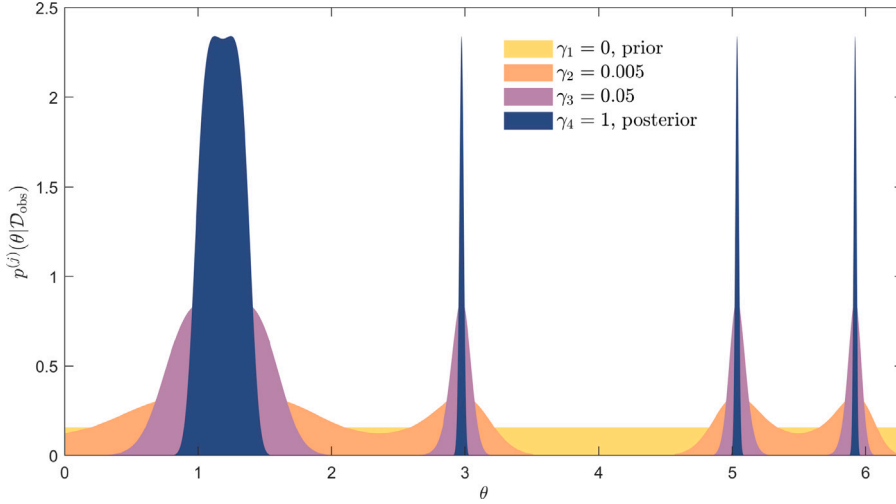


Fig. 3. Intermediate and final posterior densities defined by varying tempering parameter γ .

acquisition functions to the $(j + 1)$ th tempering stage for acquiring the most valuable training points. Fourth, design a proper **stopping criterion** for the $(j + 1)$ th stage. Fifth, develop a **sampling scheme** to generate a sample population $\mathcal{T}_{MC}^{(j+1)}$, following the mean estimation of $p^{(j+1)}(\theta|D_{obs})$ (or its unnormalized version).

To proceed with the above five key developments for the $(j + 1)$ th tempering stage, some notations need to be pre-specified. Assume N_j training points are generated to reach the stopping condition of the j th stage, resulting in a fixed GP model with mean prediction $\mu_{g,N_j}(\theta)$, and a fixed mean estimate of the likelihood $[p(D_{obs}|\theta)]^{\gamma_j}$ formulated as $\mu_{like,j}(\theta) = \exp(\gamma_j \mu_{g,N_j}(\theta))$. The resultant mean estimate of the unnormalized posterior $[p(D_{obs}|\theta)]^{\gamma_j} p(\theta)$ is then denoted as $\mu_{post,j}(\theta) = \mu_{like,N_j}(\theta) p(\theta)$. Further, given the fixed mean estimate $\mu_{Z_j} = \int_{\mathbb{T}} \mu_{like,N_j}(\theta) p(\theta) d\theta$, Eq. (24) can be reformulated as:

$$\frac{Z_{j+1}}{\mu_{Z_j}} = \int_{\mathbb{T}} \frac{[p(D_{obs}|\theta)]^{\gamma_{j+1}}}{\mu_{like,j}(\theta)} \frac{\mu_{post,j}(\theta)}{\mu_{Z_j}} d\theta. \quad (25)$$

Inspired by Eq. (25) and leveraging the sample set $\mathcal{T}_{MC}^{(j)}$ generated from the estimated posterior $\mu_{post,j}(\theta)$, the details of these five critical components are first presented, followed by a summary of the TBQ algorithm, all in sequential order.

5.2. Adaptive specification of tempering parameter

The information available for specifying/updating the tempering parameter γ_{j+1} includes the sample population $\mathcal{T}_{MC}^{(j)} = \{\theta_{MC,j}^{(k)}\}_{k=1}^{N_{MC}}$ following the fixed distribution $\mu_{post,j}(\theta)$, the up-to-date GP model $\hat{g}_N \sim \mathcal{GP}(\mu_{g,N}, \sigma_{g,N}^2)$, and a fixed value γ_j of the last stage. Intuitively, γ_{j+1} controls the divergence between the intermediate posteriors of the j th and $(j + 1)$ th stages; thus, it should be chosen to balance transitional speed and computational budget for each stage. Higher value of γ_{j+1} results in smaller number of tempering stages, but may require more model calls for a single stage.

Following TMCMC [16], γ_{j+1} is calculated based on the Coefficient of Variation (CoV) of the samples of a weight indicating the distance between the two consecutive densities $p^{(j)}$ and $p^{(j+1)}$, and also that between Z_j and Z_{j+1} . Specifically, the weight for an arbitrary sample $\theta \in \mathcal{T}_{MC}^{(j)}$ is defined as:

$$w_{j+1}(\theta) = \frac{\exp(\gamma_{j+1} \mu_{g,N}(\theta))}{\mu_{like,j}(\theta)}. \quad (26)$$

A cursory examination of Eq. (26) reveals that the variation of the samples $w_{j+1}(\theta_{MC,j}^{(k)})$ tends to increase with higher value of γ_{j+1} , given γ_j fixed. It is then reasonable to compute γ_{j+1} by minimizing the absolute difference between the CoV of $w_{j+1}(\theta_{MC,j}^{(k)})$ and ς over $\gamma_{j+1} \in (\gamma_j, 1]$. The parameter ς is suggested to be 100% for TMCMC. For TBQ, it can be relaxed to 100~200%, since the sample size N_{MC} can be specified as a higher value (e.g., several dozens of thousands) with ease, and Algorithm 1 demonstrates strong capability in learning moderately sharp posteriors. In TMCMC, γ_{j+1} is specified only once per tempering stage. In contrast, for TBQ, it is dynamically updated whenever the GP model is updated. This treatment is advantageous for controlling the distance between $p^{(j+1)}$ and $p^{(j)}$ more precisely, and also provides more flexibility for balancing exploration and exploitation.

5.3. Estimation of intermediate densities and evidences

In TMCMC, the ratios Z_{j+1}/Z_j are sequentially estimated by MC estimators based on the samples of weights, and then Z is computed by the product of these estimates. This treatment results in accumulation of errors across tempering stages as each ratio is estimated with certain prediction error, and meanwhile, the prediction accuracy of the evidence ratios may also be negatively affected by the potentially a small number of low-quality MCMC samples. This phenomenon exists in all transitional methods based on MCMC, such as subset simulation [44] and linked importance sampling [45] for structural reliability analysis. However, for TBQ, this negative effect can be eliminated. To achieve this goal, I use the description $p^{(j+1)}(\theta|D_{\text{obs}}) \propto \exp(\gamma_{j+1}g(\theta))p(\theta)$, instead of $p^{(j+1)}(\theta|D_{\text{obs}}) \propto \exp((\gamma_{j+1} - \gamma_j)g(\theta))p^{(j)}(\theta)$. This results in the mean estimation for the likelihood and the unnormalized posterior of the $(j+1)$ th stage as:

$$\mu_{\text{like},j+1}(\theta) = \exp(\gamma_{j+1}\mu_{g,N_{j+1}}(\theta)), \quad (27a)$$

and

$$\mu_{\text{post},j+1}(\theta) = \mu_{\text{like},j+1}(\theta)p(\theta), \quad (27b)$$

respectively, where N_{j+1} indicates the number of training points generated to reach the stopping condition for the $(j+1)$ th stage, and thus $\mu_{g,N_{j+1}}(\theta)$ refers to the mean prediction of the GP model trained with these N_{j+1} samples. The normalizing constant $\mu_{Z_{j+1}}$ associated with $\mu_{\text{post},j+1}(\theta)$ is given by:

$$\mu_{Z_{j+1}} = \int_{\mathbb{T}} \exp(\gamma_{j+1}\mu_{g,N_{j+1}}(\theta))p(\theta)d\theta. \quad (28)$$

The corresponding confidence bounds $[Z_{j+1,\alpha}, \bar{Z}_{j+1,\alpha}]$ are then formulated as:

$$\underline{Z}_{j+1,\alpha} = \int_{\mathbb{T}} \exp(\gamma_{j+1}(\mu_{g,N_{j+1}}(\theta) - \beta\sigma_{g,N_{j+1}}(\theta)))p(\theta)d\theta \quad (29a)$$

$$\bar{Z}_{j+1,\alpha} = \int_{\mathbb{T}} \exp(\gamma_{j+1}(\mu_{g,N_{j+1}}(\theta) + \beta\sigma_{g,N_{j+1}}(\theta)))p(\theta)d\theta. \quad (29b)$$

All the three evidence terms defined by Eqs. (28) and (29) are formulated as integral over the prior density, thus can be estimated by MC estimators formulated with a sample population, denoted as S , which is generated from the prior $p(\theta)$. The size N_{est} of S can be extremely high, e.g., $10^6 \sim 10^7$, to alleviate the variation of the MC estimators. This will not introduce much cost as the integrand is expressed in closed form, and the MC estimation requires only one implementation per tempering stage once the stage's stopping condition is satisfied, rather than during each update of the GP model. However, if required, for example in case $p^{(j+1)}$ is extremely far away from the prior, advanced schemes, such as the bridging importance sampling [45], or the harmonic mean estimator formulated with the MCMC samples [9], can also be used. For simplicity, I do not give the details as for all my tests, generating MC samples from prior is sufficient. Similarly, one can generate a CI for bounding the estimate $\mu_{\text{post},j+1}(\theta)$, which is similar to Eq. (17).

5.4. Acquisition functions

The acquisition functions should be able to inform the most valuable point for improving the prediction accuracy of $p^{(j+1)}$ and Z_{j+1} , thus the two acquisition functions, originally defined by Eqs. (19) and (20), are modified as:

$$\mathcal{A}_{\text{PUQ},N,\alpha}^{\text{I}(j+1)}(\theta) = \left[\exp(\gamma_{j+1}\underline{g}_{N,\alpha}(\theta)) - \exp(\gamma_{j+1}\bar{g}_{N,\alpha}(\theta)) \right] p(\theta), \quad (30a)$$

and

$$\mathcal{A}_{\text{PUQ},N,\alpha}^{\text{II}(j+1)}(\theta) = \exp(\gamma_{j+1}(\mu_{g,N}(\theta) + 2\beta\sigma_{g,N}(\theta)))p(\theta), \quad (30b)$$

respectively. By maximizing either $\mathcal{A}_{\text{PUQ},N,\alpha}^{\text{I}(j+1)}(\theta)$ or $\mathcal{A}_{\text{PUQ},N,\alpha}^{\text{II}(j+1)}(\theta)$, it is expected to observe the point with the highest uncertainty for predicting $p^{(j+1)}$.

5.5. Stopping condition

As mentioned in Section 5.3, it is not suggested to estimate the intermediate model evidence Z_{j+1} using the estimations of the product of the consecutive evidence ratios Z_{s+1}/Z_s ($s = 1, 2, \dots, j$). However, the posterior variation of the estimator of Z_{j+1}/Z_j can be used for controlling the prediction uncertainties associated with the $(j+1)$ th stage with low computational cost. This brings two benefits compared to using the variation of the MC estimator generated by Eq. (29). First, it allows to control the prediction uncertainties associated with only the $(j+1)$ th stage, instead of the accumulated prediction uncertainties across all previous stages. Second, it is computed with the sample population $\mathcal{T}_{\text{MC}}^{(j)}$ of the j th stage, instead of the huge one S as suggested in Section 5.3. This makes the judgment of the stopping condition numerically efficient for each update of the GP model.

Specifically, with Z_j fixed at its mean estimation μ_{Z_j} , the active credible bounds of Z_{j+1} , resulted from the sample population $\mathcal{T}_{MC}^{(j)}$ following $\mu_{\text{post},j}(\theta)$, are computed by:

$$\frac{\underline{Z}_{j+1,N,\alpha}}{\mu_{Z_j}} = \frac{1}{N_{MC}} \sum_{k=1}^{N_{MC}} \frac{\exp\left(\gamma_{j+1}\left(\mu_{g,N}\left(\theta_{MC,j}^{(k)}\right) - \beta\sigma_{g,N}\left(\theta_{MC,j}^{(k)}\right)\right)\right)}{\mu_{\text{like},j}\left(\theta_{MC,j}^{(k)}\right)}, \quad (31a)$$

and

$$\frac{\bar{Z}_{j+1,N,\alpha}}{\mu_{Z_j}} = \frac{1}{N_{MC}} \sum_{k=1}^{N_{MC}} \frac{\exp\left(\gamma_{j+1}\left(\mu_{g,N}\left(\theta_{MC,j}^{(k)}\right) + \beta\sigma_{g,N}\left(\theta_{MC,j}^{(k)}\right)\right)\right)}{\mu_{\text{like},j}\left(\theta_{MC,j}^{(k)}\right)}. \quad (31b)$$

The stopping condition for the $(j+1)$ th stage should be modified from Eq. (21) as:

$$\frac{\bar{Z}_{j+1,N,\alpha} - \underline{Z}_{j+1,N,\alpha}}{\bar{Z}_{j+1,N,\alpha} + \underline{Z}_{j+1,N,\alpha}} < \epsilon_\alpha. \quad (32)$$

One notes that, as with the estimation scheme presented in Section 5.3, the error accumulation effect is alleviated, it is then suggested to set ϵ_α as a relative high value, e.g., 0.2, for the stages with tempering parameter lower than one, and then change it to a low tolerance value, say 0.05, once the tempering parameter is calculated to be one, this way to make a better trade-off between computational cost and prediction accuracy of Z and $p(\theta|D_{\text{obs}})$.

5.6. Sampling schemes

After the stopping condition for the $(j+1)$ th tempering stage being reached, the estimation of $p^{(j+1)}(\theta|D_{\text{obs}})$ is fixed at its mean estimate. It is then required to generate a sample population $\mathcal{T}_{MC}^{(j+1)}$, following this estimated distribution, or the unnormalized version $\mu_{\text{post},j+1}(\theta)$ (see Eq. (27b)), to proceed to the training process for the next tempering stage. The first population $\mathcal{T}_{MC}^{(1)}$, which follows the prior distribution, can be generated by, e.g., simple random sampling or LHS, with ease, but for the others, a conditional sampling scheme, such as MCMC and its many variants, is required. It should be emphasized again that the target of this work is to address BMU problems with tricky posteriors. The most MCMC schemes are not suitable for directly sampling from multimodal densities due to the conflict of exploration across modes and exploitation of a single mode for choosing appropriate proposal distribution [13,15,46]. Slice sampling scheme can be applicable [47], but my implementations show that it performs well for problems with no more than three dimensions. Rejection sampling can also be a candidate choice [48], but it suffers from low acceptance rate for sharply peaked densities. In the context of my question, a favorable fact is that the target density $\mu_{\text{post},j+1}(\theta)$ admits a closed-form expression and is computationally very cheap. In response to this feature, the resampling-then-Metropolis-Hastings (R-MH) sampling scheme originally developed in the TCMC procedure is suggested, and summarized in Algorithm 2.

Principally, the R-MH algorithms consists of two critical steps: a step for resampling from $\mathcal{T}_{MC}^{(j)}$, and another step for growing Markov chains to avoid particle degradation and skip burn-in stage. For each sample $\theta_{MC,j}^{(k)} \in \mathcal{T}_{MC}^{(j)}$, a weight $w_{j+1}^{(k)} = w_{j+1}\left(\theta_{MC,j}^{(k)}\right)$ can be computed with Eq. (26) by replacing $\mu_{g,N}$ with $\mu_{g,N_{j+1}}$. Further, these weights are normalized as:

$$w_{\text{norm},j+1}^{(k)} = \frac{w_{j+1}^{(k)}}{\sum_{k=1}^{N_{MC}} w_{j+1}^{(k)}}. \quad (33)$$

Following the discrete distribution $\mathbb{P}\left(\theta = \theta_{MC,j}^{(k)}\right) = w_{\text{norm},j+1}^{(k)}$, a set of N_{MC} samples, denoted as $\mathcal{T}_{\text{cand}}^{(j+1)} = \left\{\theta_{\text{cand}}^{(k)}\right\}_{k=1}^{N_{MC}}$, can be generated randomly, with replacement, from $\mathcal{T}_{MC}^{(j)}$. Undoubtedly, $\mathcal{T}_{\text{cand}}^{(j+1)}$ follows the distribution $\mu_{\text{post},j+1}(\theta)$ with uniform weight, but may suffer from severe particle degradation, especially when ς is high. The covariance matrix of $\mathcal{T}_{\text{cand}}^{(j+1)}$ can be numerically computed as [16]:

$$\Sigma_{j+1} = \sum_{k=1}^{N_{MC}} w_{\text{norm},j+1}^{(k)} \left(\theta_{MC,j}^{(k)} - \sum_{k=1}^{N_{MC}} \theta_{MC,j}^{(k)} w_{\text{norm},j+1}^{(k)} \right)^T \left(\theta_{MC,j}^{(k)} - \sum_{k=1}^{N_{MC}} \theta_{MC,j}^{(k)} w_{\text{norm},j+1}^{(k)} \right). \quad (34)$$

To avoid particle degradation, the second step involves creating a Markov chain of length l_{chain} , with each $\theta_{\text{cand}}^{(k)}$ as a seed, using the MH algorithm, with the premise that the stationary distribution of each chain is exactly $\mu_{\text{post},j+1}(\theta)$. Specifically, with $\theta^{(k,1)} = \theta_{\text{cand}}^{(k)}$ as the initial state of the k th chain, the MH algorithm to create the $(s+1)$ th state of this chain consists of the following two steps:

- (1). Generate a sample θ^* randomly following a symmetric proposal distribution centered at $\theta^{(k,s)}$, e.g., the Gaussian distribution with covariance $\zeta^2 \Sigma_{j+1}$.
- (2). Accept θ^* as $\theta^{(k,s+1)}$ with a probability $\alpha = \min\left(1, \mu_{\text{post},j+1}(\theta^*) / \mu_{\text{post},j+1}(\theta^{(k,s)})\right)$; if rejected, let $\theta^{(k,s+1)} = \theta^{(k,s)}$.

The parameter ζ is used for scaling the variation of the proposal distribution, and is suggested to be 0.2 following Ref. [16]. The above two steps are repeated until l_{chain} states are generated for each of the N_{MC} chains. As $\mathcal{T}_{\text{cand}}^{(j+1)}$ follows exactly the distribution $\mu_{\text{post},j+1}(\theta)$ with uniform weight, the Markov chains generated with the above steps generally does not involve a long burn-in period, however, to be safe, it is advisable to set $l_{\text{chain}} = 10 \sim 30$. This will not introduce too much extra cost as $\mu_{\text{post},j+1}(\theta)$ is formulated in closed form.

With the N_{MC} chains being readily grown, the last states of all chains are ultimately collected to form the sample population $\mathcal{T}_{\text{MC}}^{(j+1)}$, which follows exactly the distribution $\mu_{\text{post},j+1}(\theta)$, and does not suffer from particle degradation, as the MH algorithm is unlikely to reject candidate samples for more than ten consecutive times. This is a big difference between the sample populations produced by TBQ and TMCMC, as, to save computational cost, only one extra state is generated for each chain in TMCMC.

Algorithm 2: The R-MH algorithm

Input: $\mu_{\text{post},j+1}(\theta)$, $\mu_{\text{post},j}(\theta)$, N_{MC} , $\mathcal{T}_{\text{MC}}^{(j)}$, ζ , l_{chain}
Output: $\mathcal{T}_{\text{MC}}^{(j+1)}$

- 1 $w_{j+1}^{(k)} \leftarrow \mu_{\text{post},j+1}(\theta_{\text{MC},j}^{(k)}) / \mu_{\text{post},j}(\theta_{\text{MC},j}^{(k)})$;
- 2 $w_{\text{norm},j+1}^{(k)} \leftarrow \text{Eq. (33)}$;
- 3 $\Sigma_{j+1} \leftarrow \text{Eq. (34)}$;
- 4 $\mathcal{T}_{\text{cand}}^{(j+1)} = \{\theta_{\text{cand}}^{(k)}\}_{k=1}^{N_{\text{MC}}} \leftarrow \text{Sample from } \mathcal{T}_{\text{MC}}^{(j)} \text{ with replacement following } \mathbb{P}(\theta = \theta_{\text{MC},j}^{(k)}) = w_{\text{norm},j+1}^{(k)}$;
- 5 $\theta_{\text{cand}}^{(k,1)} \leftarrow \theta_{\text{cand}}^{(k)}$;
- 6 **for** $k = 1, 2, \dots, N_{\text{MC}}$ **do in parallel**
- 7 **for** $s = 1, 2, \dots, l_{\text{chain}} - 1$ **do**
- 8 $\theta^* \leftarrow \text{Sample following Gaussian distribution with mean } \theta^{(k,s)} \text{ and covariance } \zeta^2 \Sigma_{j+1}$;
- 9 $r \leftarrow \min(1, \mu_{\text{post},j+1}(\theta^*) / \mu_{\text{post},j+1}(\theta^{(k,s)}))$;
- 10 $u \leftarrow \text{Randomly sample following uniform distribution within } [0, 1]$;
- 11 **if** $r \geq u$ **then**
- 12 $\theta^{(k,s+1)} \leftarrow \theta^*$;
- 13 **else**
- 14 $\theta^{(k,s+1)} \leftarrow \theta^{(k,s)}$;
- 15 **end**
- 16 **end**
- 17 $\theta_{\text{MC},j+1}^{(k)} \leftarrow \theta^{(k,l_{\text{chain}})}$;
- 18 **end**
- 19 $\mathcal{T}_{\text{MC}}^{(j+1)} \leftarrow \{\theta_{\text{MC},j+1}^{(k)}\}_{k=1}^{N_{\text{MC}}}$.

5.7. Summary of the TBQ algorithm

With all the theoretical developments in this section, the TBQ algorithm, with the name borrowed from my previous work for structural reliability analysis [49], is summarized as Algorithm 3. Compared to that of the BQ algorithm, increased complexity is mainly on the specification of the tempering parameter γ_{j+1} , which is numerically efficient as it involves solving a univariate nonlinear equation readily estimated by the MC samples. In case that the posterior show relatively small divergence from the prior, the algorithm will automatically specify $\gamma_2 = 1$, and Algorithm 3 will degrade into Algorithm 1. Therefore, users do not need to bother to make a choice of algorithm by actually understanding the behavior of the posteriors. One can just use the TBQ algorithm.

For simplicity, only the procedures for estimating the mean prediction μ_Z and $\mu_{\text{post}}(\theta)$ are shown, but one can also generate the CI predictions for these two quantities based on the details described in Section 5.3. In addition to the parameter settings of Algorithm 1, N_{MC} is suggested to be $10^4 \sim 10^5$, and N_{est} can be $10^6 \sim 10^7$, as the integrands need to be estimated with S admit closed-form expressions, and the estimations need to be performed only once per tempering stage. α is recommended to be 0.05~0.6, as already suggested in Section 4.4. Experience shows that arbitrary value of α in this range results in convergence, but may require different numbers of model calls. ϵ_α should be determined based on the value of α , and higher value of α results in higher value ϵ_α . Generally, ϵ_α is recommended to be 0.06~0.3 for intermediate stages and 0.02~0.1 for the last stage, depending on the users' tolerance to prediction uncertainties. The higher value setting for the intermediate stages is mainly benefited from the elimination of error accumulation across stages. But it should not be setting as too high as higher value results in a risk of misleading the transitional process. Besides, compared to Algorithm 1, the delayed judgment scheme has been left out, but it is encouraged. For the numerical implementations of this work, I set $N_S = 2$ steps for delayed judgment.

6. Numerical experiments and engineering applications

In this section, I introduce two numerical examples (each featuring multiple parameter settings), two dynamic examples (one nonlinear and one linear), and a real-world engineering example to demonstrate the effectiveness of the proposed methods. To

Algorithm 3: The TBQ algorithm

Input: $p(D_{\text{obs}}|\theta)$, $p(\theta)$, α , ϵ_α , ζ , N_0 , N_{MC} , N_{est}
Output: μ_Z , $\mu_{\text{post}}(\theta)$, $\mathcal{T}_{\text{MC}}^{(M)}$, N , M , $\gamma_1 \sim \gamma_M$

- 1 Initialization: $N = N_0$, $j = 0$, $\gamma_1 = 0$, $Z_1 = 1$;
- 2 $S \leftarrow$ generate N_{est} samples following $p(\theta)$;
- 3 $\mathcal{T}_{\text{MC}}^{(1)} \leftarrow$ generate N_{MC} samples from S or following $p(\theta)$;
- 4 $\mathcal{T} = \{\theta^{(k)}\}_{k=1}^N \leftarrow$ generate N_0 samples following $p(\theta)$ or from S ;
- 5 $\mathcal{Y} \leftarrow g(\mathcal{T}) = \log p(D_{\text{obs}}|\mathcal{T})$;
- 6 $D_{N,\text{train}} \leftarrow (\mathcal{T}, \mathcal{Y})$;
- 7 $\hat{g}_N(\theta) \leftarrow$ train a initial GP with $D_{N,\text{train}}$;
- 8 **while** $\gamma_{j+1} < 1$ **do**
- 9 $j \leftarrow j + 1$;
- 10 **repeat**
- 11 $\gamma_{j+1} \leftarrow$ minimize the absolute difference between the CoV of $w_{j+1}(\theta_{\text{MC},j}^{(k)})$ (Eq. (33)) and ζ over $\gamma_{j+1} \in (\gamma_j, 1]$;
- 12 $\theta^+ \leftarrow$ maximize the acquisition function formulated with Eq. (30a) or Eq. (30b);
- 13 $y^+ \leftarrow g(\theta^+) = \log p(D_{\text{obs}}|\theta^+)$;
- 14 $D_{N+1,\text{train}} \leftarrow D_{N,\text{train}} \cup \{(\theta^+, y^+)\}$;
- 15 $N = N + 1$;
- 16 $\hat{g}_N(\theta) \leftarrow$ train a GP with $D_{N,\text{train}}$;
- 17 **until** Eq. (32) satisfied;
- 18 $\mu_{Z_{j+1}} \leftarrow$ MC estimation with sample population S based on Eq. (28);
- 19 $\mu_{\text{post},j+1}(\theta) \leftarrow$ Eq. (27b);
- 20 $\mathcal{T}_{\text{MC}}^{(j+1)} \leftarrow$ Algorithm 2;
- 21 **end**
- 22 $M \leftarrow j + 1$;
- 23 $\mu_Z \leftarrow \mu_{Z_M}$, $\mu_{\text{post}}(\theta) \leftarrow \mu_{\text{post},M}(\theta)$.

compare the accuracy of results, the reference values of all (intermediate) model evidences, of the first two examples, are computed with MCS using $5 \times 10^6 \sim 7 \times 10^7$ samples to ensure that the CoV of each estimate do not exceed 0.5%.

6.1. A univariate example

The example illustrated by Fig. 1 is used here for illustrating the details of the proposed methods. The model function is formulated as $\mathcal{M}(\theta) = \theta(\sin 2\theta + 1)$. The observation is still assumed to be $y_{\text{obs}} = 2$, with the noise STD assumed to be $\sigma_n = 0.5$ (case 1) for verifying Algorithm 1, and 0.1 (case 2) for demonstrating Algorithm 3. The prior density is assumed to be uniform with support $[0, 2\pi]$, and the likelihood function is formulated with the Gaussian density as $\phi(\theta|\mathcal{M}(\theta) - y_{\text{obs}}, \sigma_n^2)$. It is unquestionably that the posterior of case 2 is more sharply peaked than that of case 1.

For case 1, the BQ algorithm is implemented with parameters setting as $N_0 = 12$, $N_{\text{MC}} = 10^5$, $\alpha = 0.2$ and $\epsilon_\alpha = 0.05$. For the final results, α is changed to 0.05 to generate the conservative CIs with credible level 95%. The results are reported in Fig. 4, where (a) and (b) refer to the 95% CI estimates of the posterior density generated by the two PUQ acquisition functions respectively, together with the true results as reference, as well as the initial and actively designed training points. Initialized with the same set of $N_0 = 12$ training points, each acquisition function sequentially produces 12 more training points to reach the stopping condition, thus the costs of the two implementations are the same. It is also seen that the BQ algorithm with either acquisition function produces accurate estimation of the posterior, and all the four modes are captured with very small numerical error. Comparison of Fig. 4(a) and (b) indicates that the adaptive training points (indicated by the stars) are all located in the supports of the four modes, demonstrating the effectiveness of the two acquisition function for observing the true modes. This is definitely benefited from the exponential impact, as explained in Section 4.1.

During each training step, the 95% CIs are computed for model evidence, and the evolution of the CIs' widths against the iteration step is reported in 4(c). It is empirically shown that the two acquisition function shows similar convergence rate for this case. It is hard to say which acquisition function is preferable. Thus, only the first type of acquisition function is utilized for TBQ for all the remaining examples. Interested readers are encouraged to conduct further tests on the two acquisition functions using the openly accessible code (link provided in the conclusion section).

The TBQ algorithm is then implemented to estimate the sharply peaked posterior of case 2. The algorithm parameters are set to be $N_0 = 12$, $\epsilon_\alpha = 0.05$ for intermediate stages and 0.02 for the final stage, $N_{\text{MC}} = 10^4$, $N_{\text{est}} = 5 \times 10^6$, $\zeta = 1.2$, and $\alpha = 0.2$. The results of all tempering stages are reported in Fig. 5, including the mean estimate, the 95% CIs and the reference (intermediate) posteriors. The tempering parameters γ_j automatically specified by the algorithm as well as the number of accumulated model calls are also reported. It is seen that, by initializing γ_1 as 0, the algorithm sequentially produced three intermediate posterior densities, with the

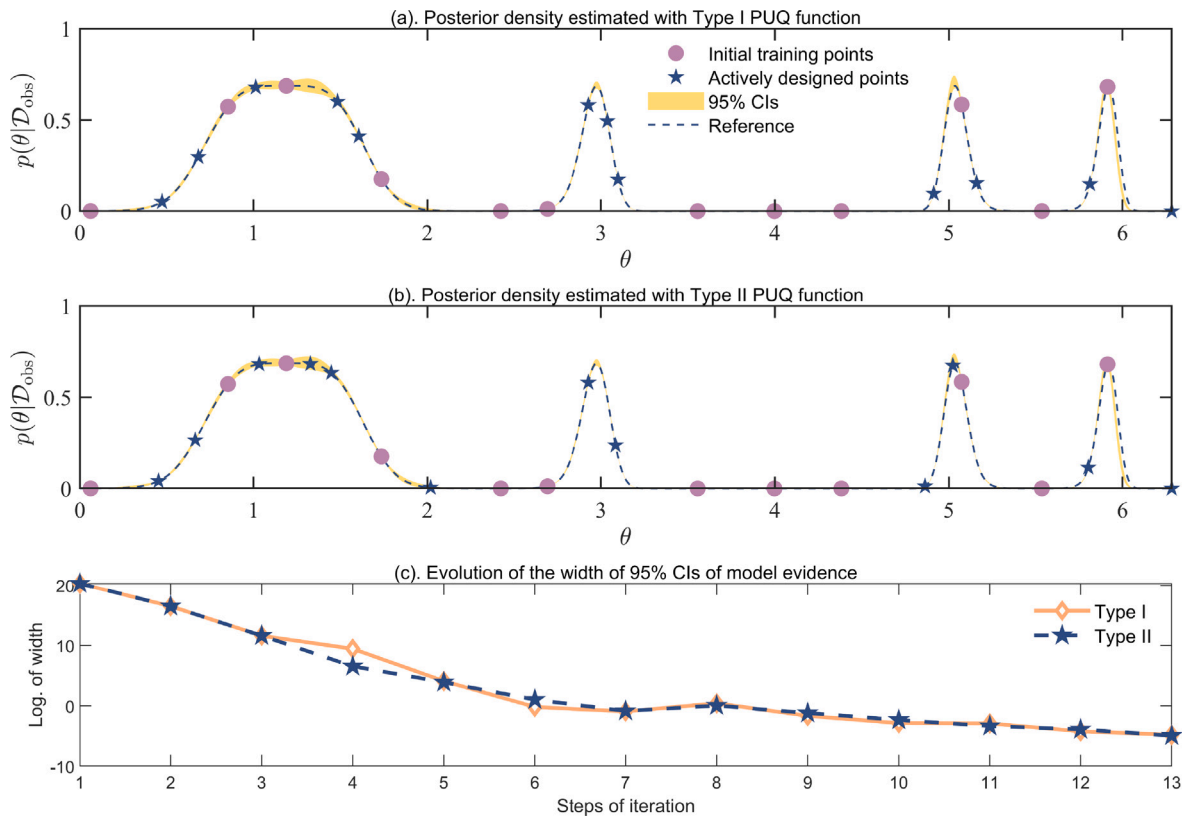


Fig. 4. Results for case 1 of the 1D example, with (a) referring to the results generated by the first type of PUQ function, (b) showing the results generated by the second type of PUQ function, and (c) referring to the evolution of the 95% CIs of model evidence over the steps of iteration.

last one being the true posterior as $\gamma_4 = 1$. It is seen that the mean estimates of all the three (intermediate) posterior densities match exactly with the references solutions and the 95% CIs are extremely narrow, indicating the extremely high accuracy of the TBQ for each intermediate stage. However, this can be relaxed to save computational cost as the error accumulation effect is eliminated for TBQ. From the next example on, the requirement of accuracy for intermediate stages will be relaxed. Here, the error tolerance is set to be 0.05 for all intermediate tempering stages for illustrating the rationale, flexibility and potentials of the TBQ algorithm.

Fig. 5 also demonstrates that, initialized with 12 training samples randomly drawn by LHS, 11 more training points are actively produced for accurate estimation of the first intermediate posterior density, and for each of the subsequent stage, only three extra training points are required, resulting in a total number of 29 model calls for approaching the target posterior. It can also be observed that all the actively added training points are located within the supports of the (intermediate) posteriors, demonstrating the high effectiveness of the acquisition functions for designing training points.

To mitigate the effect of initial training points and demonstrate the robustness of the TBQ algorithm, it is executed ten times under identical parameter settings for case 2, and the results of γ_j as well as Z_j are reported in Fig. 6(a) and (b) respectively, together with the number of consumed model calls and associated numerical errors (measured by the widths of the 95% CIs). It is seen, that all the replications produce three tempering stages, but the values of γ_j of the same level may differ across different implementations. It is also seen from 6(b) that, with each of these ten replications, the mean estimates (implied by bars) of all Z_j are in good agreement with their reference values (indicated by the diamond signs), and the widths of the 95% CIs are sufficiently small compared to the values of Z_j , indicating the high accuracy of all these ten replications.

Finally, the influence of parameter settings of the TBQ algorithm on its performance is investigated. Although there are a number of parameters, ζ and ϵ_α are the two most critical ones for the algorithm, and thus are specifically investigated. By setting ζ as 0.5, 1.2, and 2, and the other parameters set as the same as stated above, the results of the (intermediate) posterior densities and model evidence produced by TBQ are reported in Fig. 7. These three implementations are initialized with the same set of training points created with LHS. It is clearly seen that 9, 3 and 2 tempering stages have been produced by setting ζ as 0.5, 1.2 and 2 respectively. Despite the difference, all the three implementations have produced accurate estimates of the posterior density, with all the four modes having been captured with high accuracy. Besides, the three settings almost consumed the same number of model calls, and it is hard to say which is better. Overall, smaller value of ζ results in more intermediate tempering stages.

Further, by setting ϵ_α as 0.1, 0.05, 0.01 for all tempering stages, and then the other parameters as the same as that for generating Fig. 5, the results of (intermediate) posterior densities as well as the model evidence are reported in Fig. 8. It is evident that all these

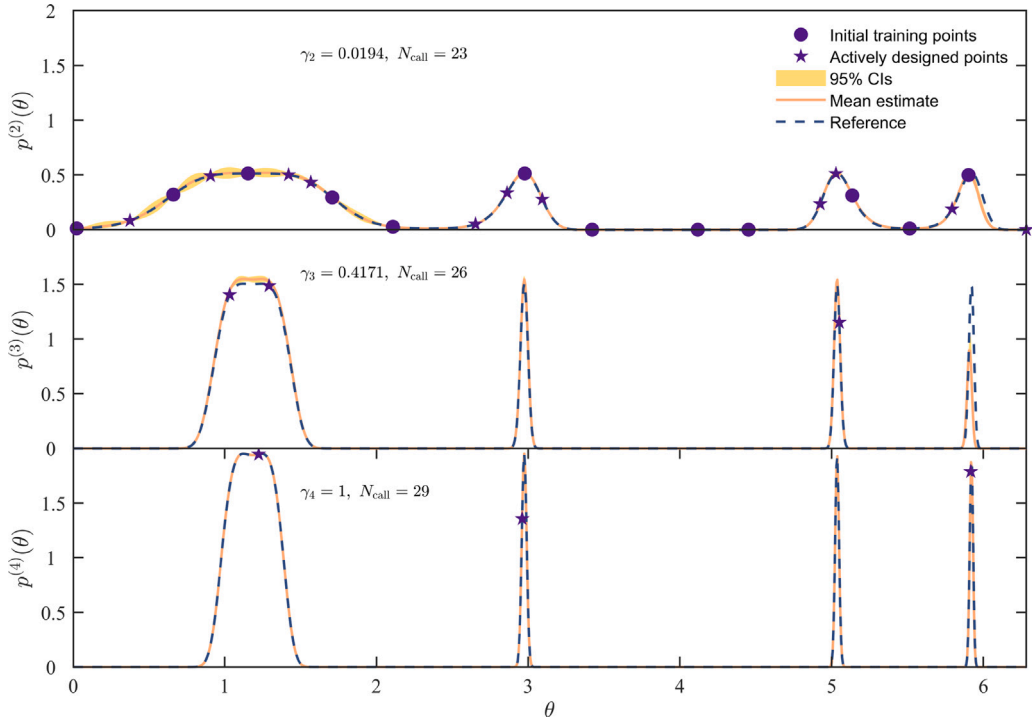


Fig. 5. Results of TBQ for case 2 of the univariate example. The three subplots sequentially show the 95% CIs of the posteriors for the three intermediate stages, with the values of γ_j and the accumulated number N_{call} of functions calls labeled in the corresponding subgraphs. The training points generated for each stage, as well as the reference posterior of each intermediate stage are also reported for comparison.

three runs generated three intermediate tempering stages and estimated the posterior density with high precision. As ϵ_α increases, the number of required model calls decreases, while the CIs of the model evidence becomes wider. Therefore, the parameter ϵ_α serves to balance the trade-off between estimation accuracy and computational efficiency. Overall, results in Figs. 7 and 8 demonstrate that the performance of the TBQ algorithm is robust to both parameters ζ and ϵ_α .

6.2. A multi-dimensional multi-modal (MDMM) example

The aim of this example is to verify the effectiveness of the TBQ algorithms for estimating low to medium dimensional issues with complex posterior configurations, including multiple and sharp modes as well as linear and nonlinear dependencies. To this aim, three energy functions are defined as follows:

$$\mathcal{U}_1(\theta|\sigma_2, \sigma_1) = \frac{1}{2} \left(\frac{\|\theta\| - 2}{\sigma_2} \right)^2 - \log \left(\exp \left(-\frac{1}{2} \left(\frac{\theta_1 - 2}{\sigma_1} \right)^2 \right) + \exp \left(-\frac{1}{2} \left(\frac{\theta_1 + 2}{\sigma_1} \right)^2 \right) \right) \quad (35a)$$

$$\mathcal{U}_2(\theta|\sigma_3, \sigma_4, \rho) = \frac{1}{2} (\theta - \mu_\theta)^\top \begin{pmatrix} \sigma_3^2 & \rho \sigma_3 \sigma_4 \\ \rho \sigma_3 \sigma_4 & \sigma_4^2 \end{pmatrix}^{-1} (\theta - \mu_\theta) \quad (35b)$$

$$\mathcal{U}_3(\theta|\sigma_5) = \frac{1}{2} \left(\frac{\theta_6 - \sin \left(\frac{\pi}{4} \theta_5 \right)}{\sigma_5} \right)^2, \quad (35c)$$

where $\mathcal{U}_1$ and $\mathcal{U}_3$ are adapted from Ref. [42], and for $\mathcal{U}_2$, $\mu_\theta = (0.5, 1)^\top$. The six parameters $\sigma_1 \sim \sigma_5$ and ρ are used for controlling the sharpness of modes.

With these three energy functions as negative logarithm of likelihood, $\mathcal{U}_1$, $\mathcal{U}_2$ and $\mathcal{U}_3$ produce disconnected modes, linear dependence and nonlinear dependence, respectively, for the posterior. The likelihood function is then defined as:

$$p(D_{\text{obs}}|\theta) \propto \exp(-\mathcal{U}(\theta)), \quad (36)$$

where $\mathcal{U}(\theta)$ is a linear combination of the three energy functions. Three cases are designed for verification of the TBQ methods, with the composite energy functions and the associated parameter values defined in Table 2. For the second and third cases, ten model parameters $\theta_1 \sim \theta_{10}$ are considered for calibration, but some (six for case 2 and four for case 3) are defined as non-influential

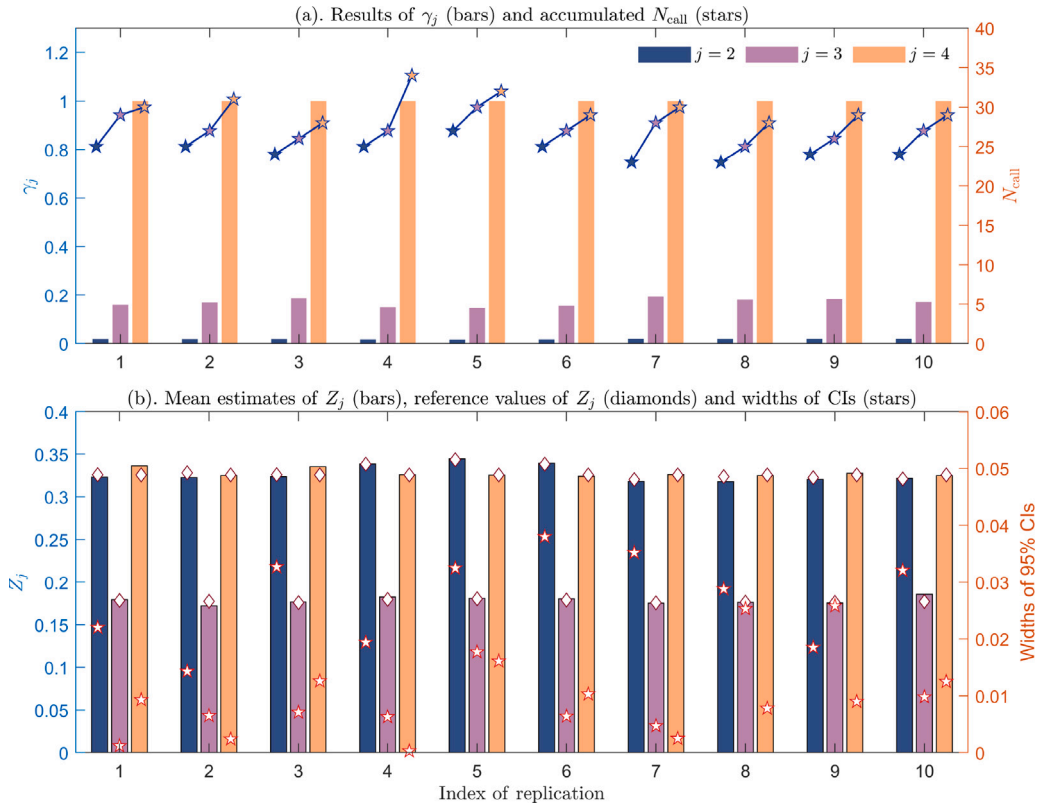


Fig. 6. Results of γ_j and N_{call} (a) as well as Z_j (b) for case 2 of the 1D example, generated by implementing TBQ repeatedly for ten times, where the reference values of Z_j are computed with MCS using 10^6 samples.

Table 2

Three different settings of energy functions $\mathcal{U}(\theta)$ for the MDMM example.

Index	$\mathcal{U}(\theta)$	Parameter values	d
Case 1	$\mathcal{U} = \mathcal{U}_1(\theta_1, \theta_2)$	$\sigma_1 = 0.08, \sigma_2 = 0.2$	2
Case 2	$\mathcal{U} = \mathcal{U}_1(\theta_1, \theta_2) + \mathcal{U}_2(\theta_3, \theta_4)$	$\sigma_3 = 0.7, \sigma_4 = 1, \rho = 0.8$	10
Case 3	$\mathcal{U} = \mathcal{U}_1(\theta_1, \theta_2) + 0.3\mathcal{U}_2(\theta_3, \theta_4) + \mathcal{U}_3(\theta_5, \theta_6)$	$\sigma_5 = 0.3$	10

parameters with the marginal posterior equals to the marginal prior, this way to check if the TBQ algorithm is capable of successfully identify the non-influential parameters. Besides, given the closed-form expressions of the normalized posterior density for each pair of parameters, it is possible to plot the corresponding exact marginal posterior densities as references to check the quality of numerical estimations.

For case 1, the algorithm parameters are set as $N_0 = 12$, $N_{\text{MC}} = 10^4$, $N_{\text{est}} = 5 \times 10^6$, $\zeta = 1.2$ and $\alpha = 0.6$. ϵ_a is set to be 0.15 for intermediate tempering stages, and then 0.03 for the last stage. Under this setting, the TBQ algorithm is implemented, and results of the intermediate posterior densities for all tempering stages are reported in Fig. 9(a–d). The algorithm automatically produces four tempering stages, with the last one being the estimation of the target posterior. The true target posterior density is reported in Fig. 9(e) for comparison, and the absolute difference between the densities in Fig. 9(d) and (e) is shown in the last panel. It can be seen that the mean estimate generated with TBQ match well with the reference posterior, and the absolute error is sufficiently small, indicating the high accuracy for estimating the posterior density. Besides, the mean estimate of Z corresponding to the estimate in 9(d) is 0.0120, with the CI being [0.0117, 0.0123], and the reference value being 0.0120, indicating that the model evidence is also estimated with high accuracy. Initialized with $N_0 = 12$ random training samples, TBQ automatically produces 7, 5, 4, and 4 training samples for the four sequential stages respectively, results in a total number of 32 calls of the likelihood function. It is also seen that, for each tempering stage, the actively designed training points are almost all located within the supports of the modes of the corresponding intermediate posteriors, demonstrating the high effectiveness of the proposed PUQ acquisition function.

Additionally, HMC [15], Variational Bayesian quadrature (VBQ) [18] and TMCMC [16] are also implemented to generate the posterior samples for case 1, and the results are compared in Fig. 10. For HMC, the step size parameter is set to be 0.02, the number of leapfrog steps is assumed to be 100, and the burn-in sample size is specified as 100; For VBQ, the initial training sample size is set to be $10[(d+1)/10]$, with $\lceil \cdot \rceil$ being the ceiling function, the threshold for stopping judgment is to assumed to be 0.1, and

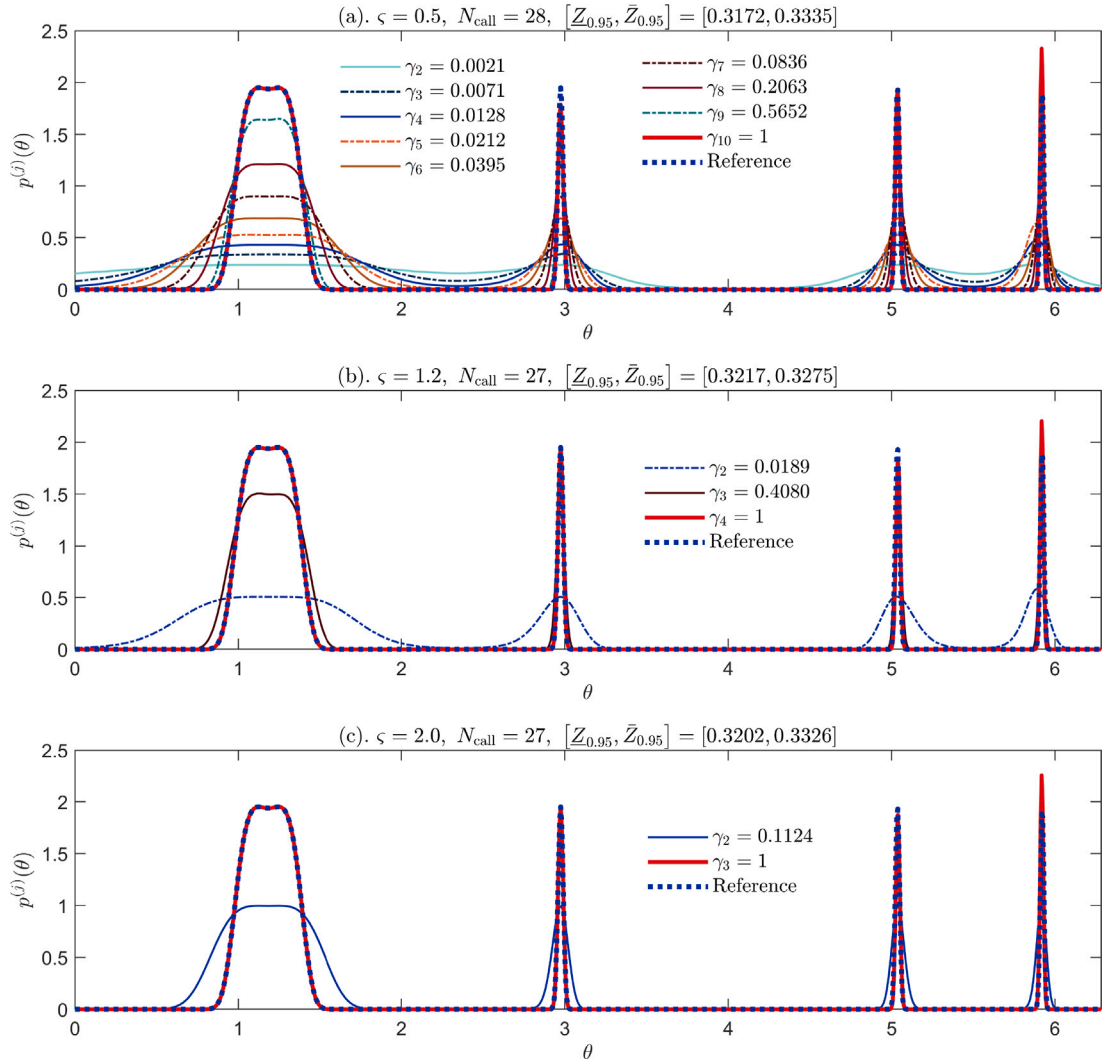


Fig. 7. Results of (intermediate) posterior densities for case 2 of example 1, generated by setting ζ as 0.5, 1.2 and 2 respectively.

the number of added points in each iteration step is 5; For TMCMC, the parameter ζ is assumed to be 100%, and the sample size for each stage is specified to be 10^4 . The above parameter settings apply to all the test examples in this work, and one can refer to the respective references for details of these parameters. It is seen that, both VBQ and TMCMC successfully captured both modes with high accuracy, while HMC captures only one mode. For these implementations, HMC consumed 2.02×10^4 calls of the model function; VBQ used 90 model calls; and TMCMC made 5.6×10^5 calls to the model. This further proves the high efficiency of TBQ.

To further validate the algorithm's robustness, it is executed five times under identical parameter settings, each initialized with a random set of 12 training samples. The training details and the estimation of model evidences for each tempering stage are reported in Fig. 11. It is seen from 11(a) that all the five implementations produce four tempering stages, but the identified values of γ_j for each fixed j show bigger variation, compared to those reported in Fig. 6. This is mainly caused by the relaxed stopping thresholds for the intermediate stages. Despite this difference, all implementations produce high quality of estimation for the target model evidence, as indicated by 11(b). This demonstrates again the theoretical arguments that the error accumulation across tempering stages has been avoided. This brings benefits not only for achieving high accuracy, but also for saving computational cost, as under this setting, the intermediate tempering stages are not necessarily estimated with very low error tolerance. It only needs to be coarsely specified to ensure that the learning capacity required for each stage does not exceed the BQ algorithm's capability. Fig. 11(c) shows that the total numbers of model calls consumed by these five implementations are within the range of 32 and 37, indicating the robustness of the algorithm in terms of efficiency.

For case 2, ζ_a is relaxed to be 0.3 for the intermediate stages, and then changed to 0.03 once $\gamma = 1$ is reached. ζ is set as 1.8. The results of one implementation are reported in Fig. 12. Although the joint posterior of the ten parameters are estimated with a closed-form expressions, it is easier to use the samples created for each stage to show the result, as reported in the first four

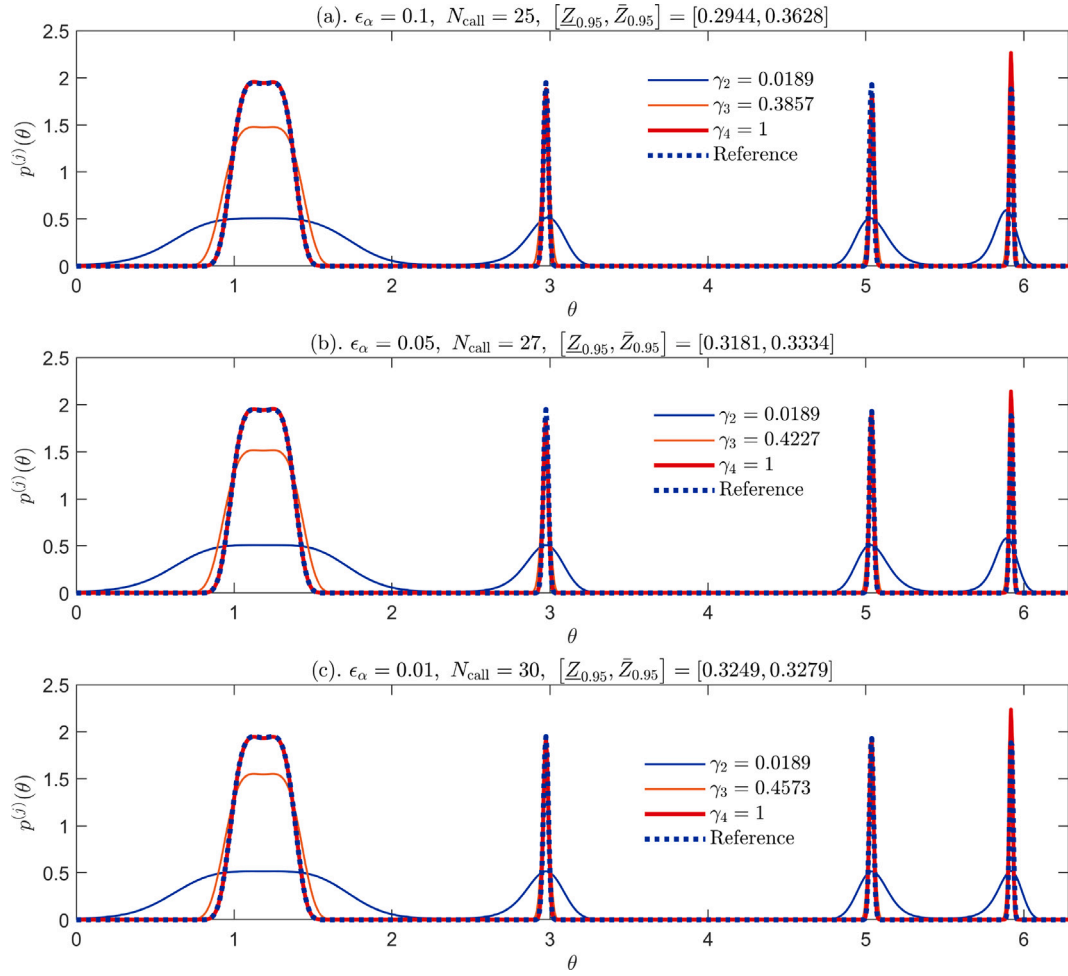


Fig. 8. Results of (intermediate) posterior densities for case 2 of example 1, generated by setting ϵ_α as 0.1, 0.05 and 0.01, respectively, for all tempering stages.

column of Fig. 12, for which the corresponding values of γ_j are reported in the top of the first row. It has been explained that as the ten parameters are dependent in pairs, thus the samples are shown in pairs, with the corresponding reference target unnormalized marginal posteriors compared in the last column. As $\theta_5 \sim \theta_{10}$ are all set as non-influential, only the results of (θ_5, θ_6) are shown, for simplicity. It is clearly seen by the last two columns of Fig. 12 that, the multimodal and sharp dependence of (θ_1, θ_2) , the linear dependence of (θ_3, θ_4) , and the negligibility of (θ_5, θ_6) , have all been identified with high accuracy. It is also seen that, for each intermediate stage, alternative types of patterns have all been correctly identified by the produced samples as they all cover the patterns of the subsequent stages. The mean estimate of the model evidence corresponding to the results in Fig. 12 is 4.793×10^{-4} with a CI $[4.7277, 5.1250] \times 10^{-4}$. The reference solution computed with MCS using 5×10^7 samples is 4.9620×10^{-4} , with the CoV being 0.34%. This sufficiently demonstrated the high accuracy for estimating the model evidence. The corresponding number of model calls is 83, implying the high efficiency. Through comparing the fourth and fifth columns of Fig. 12, one may perceive that the supports of the estimated posteriors are slightly larger than those of the reference posteriors. However, this is not the case, the supports of each pair are almost identical. If it is the case, the estimated model evidence will be smaller than its reference value, as higher variation of posterior commonly corresponds to smaller model evidence. As already demonstrated, the estimated model evidence matches well with its reference.

The posterior samples generated by HMC, VBQ and TMCMC are reported in Fig. 13. It is clearly seen that the linear dependent pattern of θ_3 and θ_4 as well as the non-influential pattern of θ_5 and θ_6 are both correctly captured by the three methods. However, for the multimodal pattern of θ_1 and θ_2 , only the TMCMC shows high recognition accuracy, and only one mode out of the two is captured by either of the other two methods. Indeed, the high performance of TBQ for this problem is adapted from TMCMC, but definitely it consumed much less model calls. For these implementations, HMC consumed the same number of calls to the model as that for case 1, while VBC and TMCMC respectively made 235 and 8.4×10^5 calls to the model, indicating the higher efficiency of TBQ.

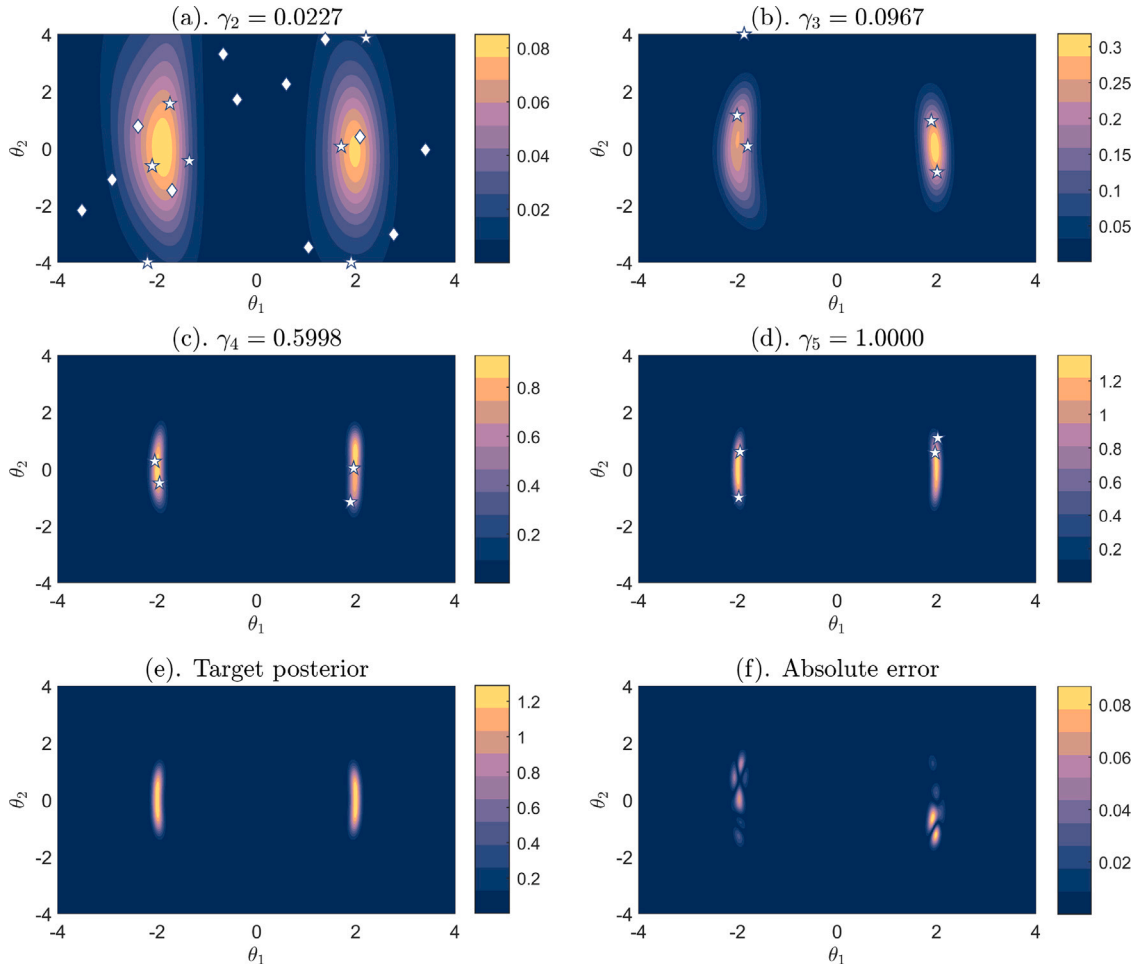


Fig. 9. Results of intermediate posterior densities for case 1 of the MDMM example (a-d), each of which is labeled with the adaptively specified value of γ_j , together with the true posterior (e) for comparison, the absolute error (f) for illustrating the accuracy, the initial training points (diamonds), as well as the training points actively generated during each iteration (stars).

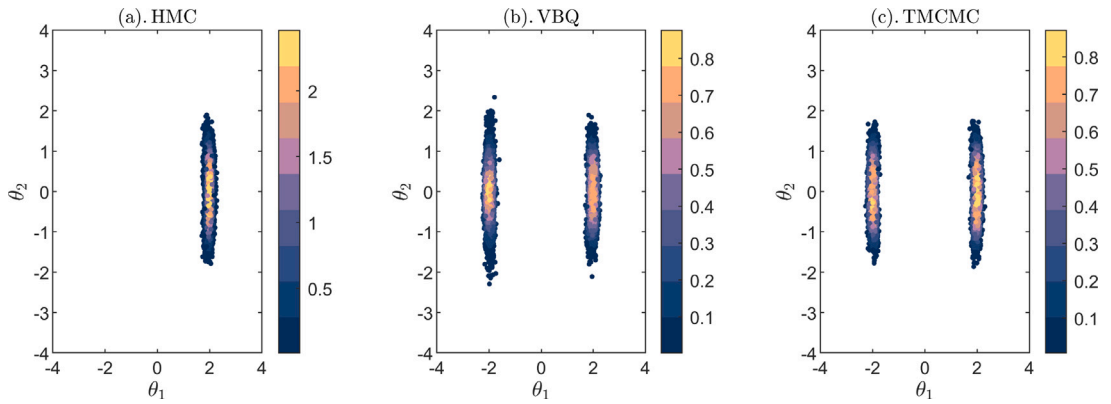


Fig. 10. Posterior samples generated by HMC, VBQ and TMCMC for case 1 of the MDMM example.

To further verify the robustness of the algorithm for case 2, results of γ_j , $\log Z_j$ and N_{call} generated with five implementations are reported in Fig. 14. It is seen that, all implementations produce four tempering stages, and all produce accurate estimation of the model evidence for the final stage, as revealed by Fig. 14(b). The total numbers of model calls for these five replications, to reach

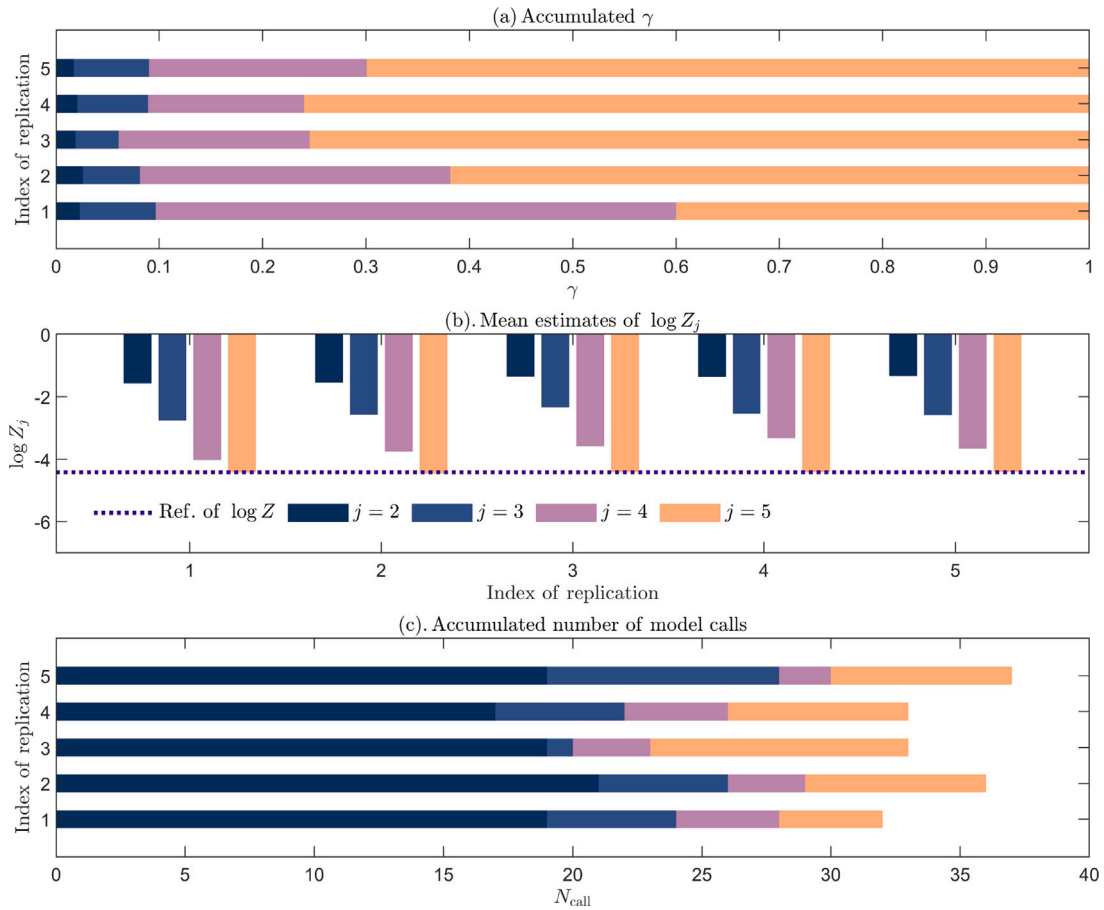


Fig. 11. Results of five replications for case 1 of the MDMM example, where (a) shows the accumulated γ values for the sequential tempering stages, (b) indicates the results of the logarithm of the model evidences for all tempering stages, and (c) displays the accumulated number of model calls.

the same stopping condition, vary from 83 to 102. The randomness of the numbers of model calls across these five runs are mainly caused by the randomness of the initial training samples. Despite this randomness, the TBQ algorithm exhibits high robustness in terms of prediction accuracy. The above facts demonstrate the robustness of the TBQ algorithm, in terms of both accuracy and efficiency, for this medium-dimensional challenging case.

On the basis of case 2, a severely nonlinear dependence pattern is introduced in case 3, leading to a more complex posterior structure and greater divergence between posterior and prior. This poses a significantly greater challenge. For this case, the stopping threshold ϵ_a is set to be 0.25 for stages with $\gamma < 1$, and 0.08 for the final stage. The sample size N_{MC} is set to be 10^4 , and N_{cand} is set to be 2×10^7 . ζ is set to be 1.2. This implementation of TBQ automatically produced eleven tempering stages, and for simplicity, results of five intermediate stages are reported in Fig. 15, together with the target densities for comparison. The numbers of function calls for these 11 stages are, in turn, 44, 4, 2, 8, 12, 34, 9, 12, 102, 9, and 96, summing to a total number of 332 calls. The mean estimate of the target model evidence is 1.2655×10^{-4} , with the CI being $[1.1021, 1.4664] \times 10^{-4}$. The reference solution computed by MCS is 1.4593×10^{-4} , with a CoV being 0.46%. Thus, the model evidence is estimated with desired accuracy. The last two columns of this figure show that all the complex topology of posterior, including the multiple distant and disconnected modes, linear and nonlinear dependencies, as well the negligibility have been pinpointed. These results have significantly proved the effectiveness of the TBQ algorithm for BMU with extremely tricky posteriors.

The posterior samples for case 3 generated by HMC, VBQ and TMCMC are shown in Fig. 16. It is seen again, that TMCMC successfully captured all the four types of patterns, i.e., the multimodality, the linear dependence, the nonlinear dependence and the insignificance. HMC failed to capture the multimodality, but successfully identified the other three types of pattern. Comparably, VBQ is less accurate than the other three methods for this case. For this case, HMC consumed the same number of calls to model, while VBQ and TMCMC respectively made 385 and 8.4×10^5 calls to the simulator. This has sufficiently demonstrated the superiority of TBQ over the three methods.

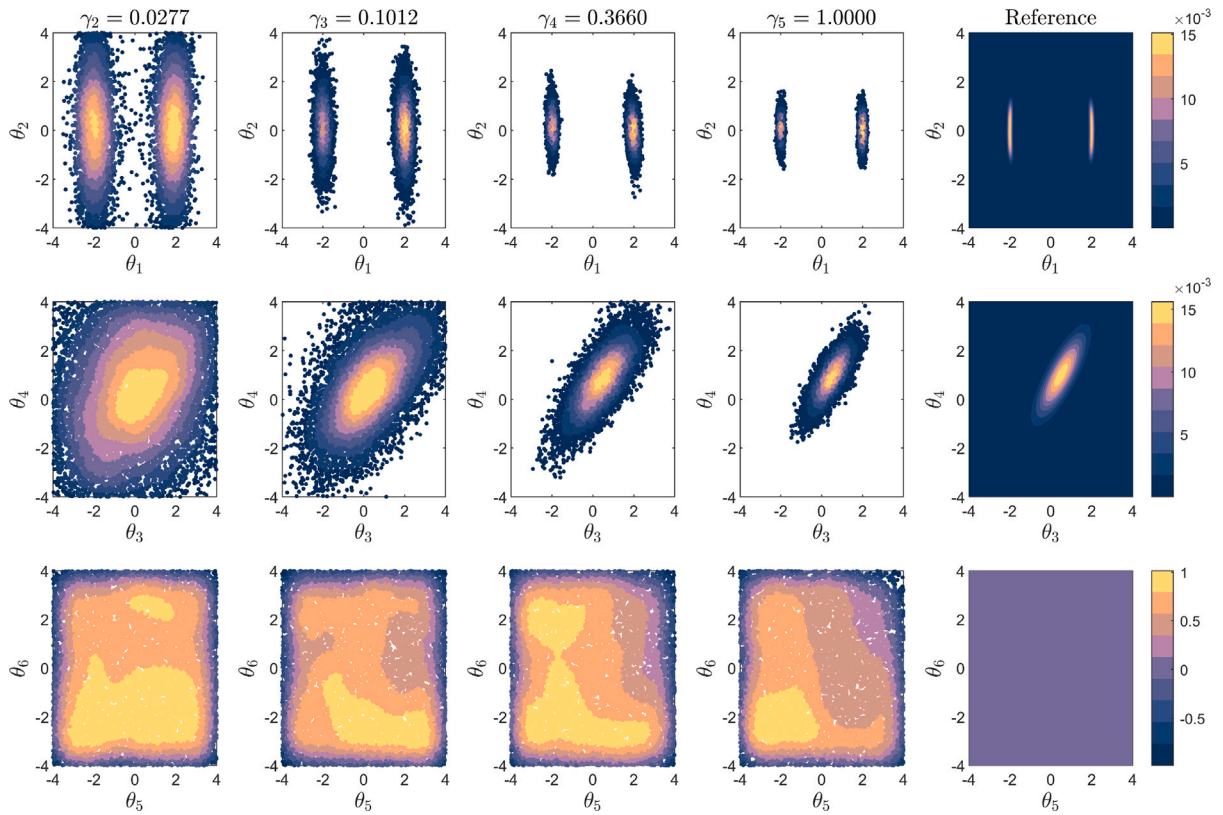


Fig. 12. Results of posterior densities for case 2 of the MDMM example, where the first four columns display samples for each intermediate stage generated by TBQ, and the last column shows the true target densities (unnormalized). All plots correspond to pairs of dependent parameters, with brighter colors indicating higher sample density. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

6.3. A nonlinear dynamic model

Consider a simple physical system consists of planar pendulum with a periodic rotational base excitation $\Omega(t) = \Omega_0 \sin(\omega t)$. Due to nonlinear interactions, the system show dual-stable amplitude–frequency characteristics. The governing equation of this dynamic system is given as:

$$y'' + c y' |y'| + \sin(y) = m l \Omega_0 \omega \cos(\omega t) \cos(y), \quad (37)$$

where c refers to the damping coefficient, m and l are the mass and length of the planar pendulum respectively, $g = 9.8 \text{ m/s}^2$ is the gravitational acceleration. Due to the velocity-squared damping mechanisms characterized by $c y' |y'|$, the time-dependent property of the stiffness caused by the nonlinear excitation $m l \Omega_0 \omega \cos(\omega t) \cos(y)$, and the nonlinear term $\sin(y)$ representing the geometric nonlinearity, the amplitude–frequency response show a jump phenomenon. The numerical model is obtained by solving Eq. (37) using the Runge–Kutta algorithm with variable step sizes. The model prediction, as well as the quantity being observed is the amplitude. The controllable input $x = \Omega_0$ is the amplitude of the driving force, and the three model parameters to be inferred include the mass m , the damping coefficient c and the frequency ω of the excitation, that is $\theta = (m, c, \omega)^T$. The length l of the planar is fixed at 1 m. Three measurements observed at $\Omega_0 = 1, 2$, and 3 are 0.2276 rad, 0.4446 rad and 0.6429 rad respectively, with a Gaussian noise of STD 0.05. The likelihood is then formulated using the Gaussian density following Eq. (2). The prior distribution of m , c and ω are all assumed to be uniform with supports $[1.5, 1.6] \text{ kg}$, $[0.1, 1] \text{ N m s/rad}$ and $[1, 5] \text{ rad/s}$ respectively. The reference value of the three parameters are $m = 1.5$, $c = 0.2$ and $\omega = 2$ respectively.

The TBQ algorithm parameters are set to be the same as case 1 of the MDMM example, except that ζ is set to be 1.8. Initialized with $N_0 = 12$ random points, the TBQ algorithm produces two intermediate stages with the tempering parameters sequentially being 0.1933 and 1, and a total number of 51 model calls are consumed to reach the finals stopping condition. The results generated with the TBQ algorithm are reported in Fig. 17, together with the reference results generated by TMCMC. As for TMCMC, ζ is set to be 1, and the sample size of each stage is set to be 10^4 samples. A total number of three tempering stages are produced, and thus the total number of model calls is 3×10^4 .

It is shown by Fig. 17 that the posterior densities estimated by the two methods match well, indicating the high accuracy and efficiency of the TBQ method. It is seen that, among the three parameters, only the excitation frequency ω show high influence,

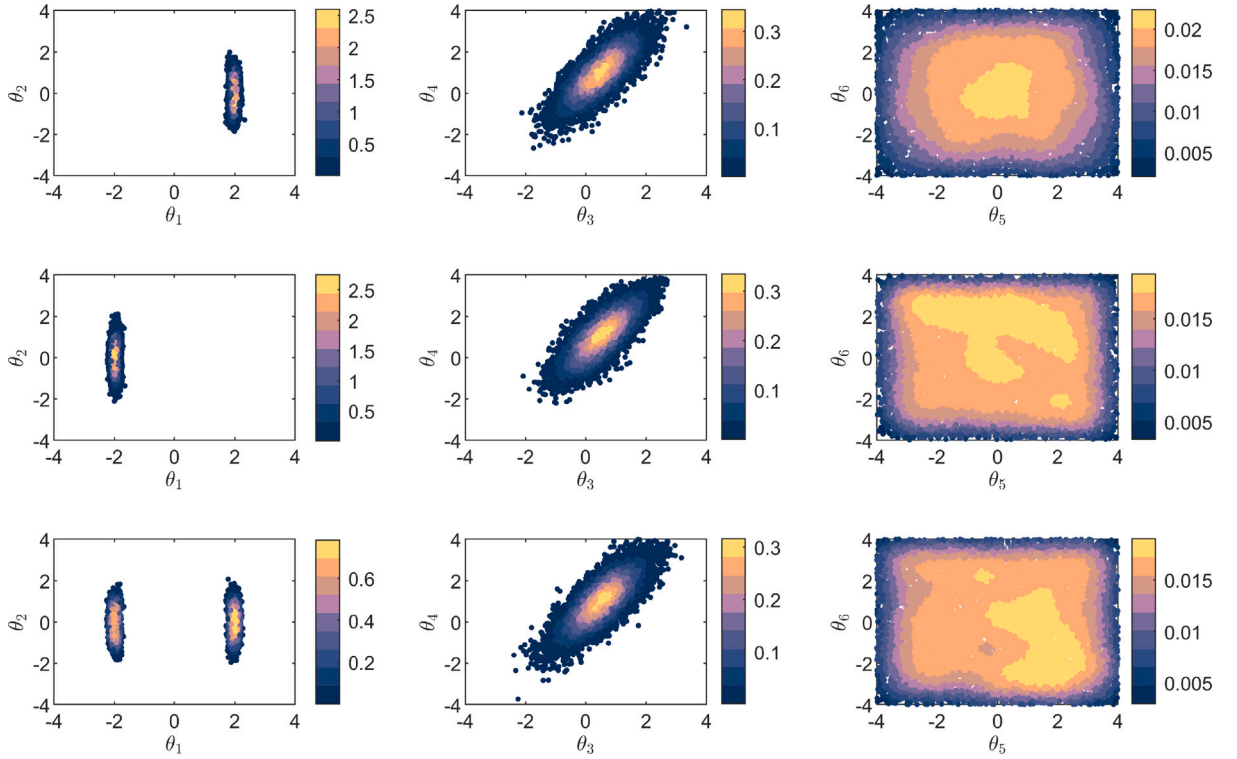


Fig. 13. Posterior samples generated by HMC (first row), VBQ (second row) and TMCMC (third row) for case 2 of the MDMM example.

and its posterior show big difference with its prior. For the first parameter, it seems that the result of TMCMC is sharper, but this is actually due to small estimation error. Indeed, the first two parameters are both non-influential, indicating that their posteriors are close to their priors, as indicated by the figure. It is also seen that the posterior density covers the true value $\omega = 2$ with high probability, further indicating the correctness of the method. Fig. 17 show that, although both methods produced 10^4 posterior samples, the samples produced by TMCMC appear significantly fewer than those created by TBQ, indicating that TMCMC results exhibit a certain degree of particle degeneracy, whereas, TBQ does not. It is also shown that the posterior of ω has two disconnected modes, which is consistent with the results of amplitude–frequency analysis which presents a trend of increasing first and then decreasing.

The mean estimate of the model evidence by TBQ is 48.4768 with the CI being [47.1017, 49.9208], and the mean estimate by TMCMC is 50.4034, indicating that the model evidence is also estimated by TBQ with sufficient accuracy. Ultimately, under identical parameter setting, TBQ is repeated ten times. On average, it consumed 66.1 model calls to reach the stopping condition, with the average mean estimate of model evidence being 48.4372 and 1.51% CoV across the ten estimates. These results further reveal that while tempering parameter values and model call counts vary across runs, each execution yields accurate estimate of both posterior densities and model evidence, demonstrating the high robustness of TBQ.

6.4. A three-degree-of-freedom (3DoF) dynamical system

Consider the dynamical analysis of a 3DoF undamped spring-mass system illustrated in Fig. 18. The governing PDE is given as $\mathbf{M}\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{0}$, where $\mathbf{M} = \text{diag}(m_1, m_2, m_3)$ refers to the mass matrix and

$$\mathbf{K} = \begin{pmatrix} k_1 & -k_1 & 0 \\ -k_1 & k_1 + k_2 & -k_2 \\ 0 & -k_2 & k_2 + k_3 \end{pmatrix} \quad (38)$$

being the stiffness matrix. The three natural frequencies are measured to be 11.9191, 26.7954 and 56.5167 respectively, with a Gaussian white measurement noise whose STD is 0.05. The masses are assumed to be known as $m_1 = 1000$ kg, $m_2 = 1200$ kg and $m_3 = 800$ kg. The stiffness of three springs are respectively $k_1 = \bar{k}\theta_1$, $k_2 = \bar{k}\theta_2$ and $k_3 = \bar{k}\theta_3$, where $\bar{k} = 21.6 \times 10^6$ N m is the nominal stiffness, and $\theta_1 \sim \theta_3$ are the three parameters to be calibrated. The prior distribution of these three parameters are all assumed to be uniform with support $\theta_1 \sim [0.1, 1]$, $\theta_2 \sim [0.5, 5]$ and $\theta_3 \sim [0.5, 5]$. The likelihood function is defined with Eq. (2).

The TMCMC method is implemented, with $\varsigma = 1$ and $N_{\text{MC}} = 4 \times 10^4$, to create reference estimates for both posterior and model evidence. Due to the sharply peaked feature of the posterior, the method adaptively produced nine tempering stages, and the model

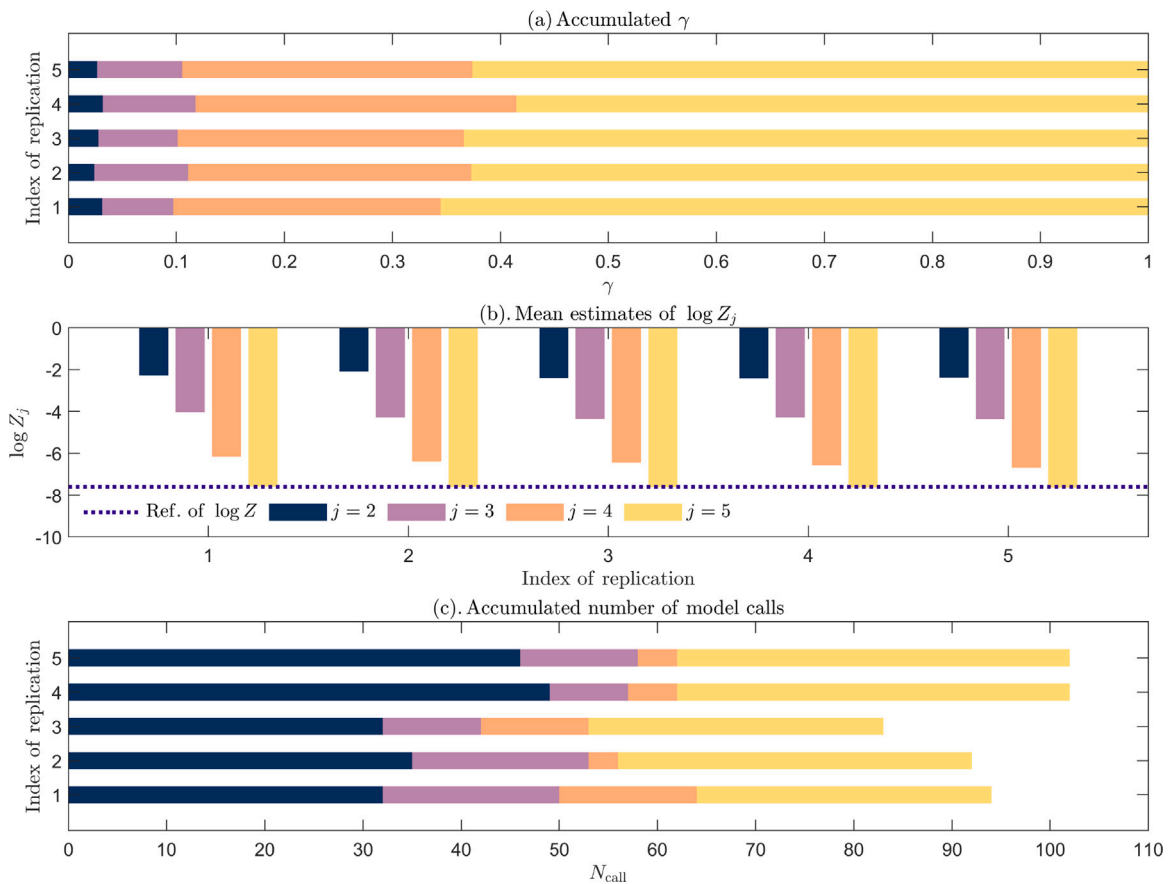


Fig. 14. Results generated with five implementations of TBQ for case 2 of the MDMM example, with (a) being the results of γ_j , (b) showing the mean estimate of $\log Z_j$, and (c) presenting the accumulated numbers of model calls.

evidence is estimated to be 7.8202×10^{-4} , and the posterior samples schematically shown in Fig. 19. For TBQ, $N_0 = 12$, $\varsigma = 2$, $N_{\text{MC}} = 10^4$, and the stopping threshold ϵ_a set to be 0.06 for intermediate stages, and then 0.03 for the final stage. Under this setting, TBQ automatically produced six tempering stages to approach the target posterior, with the total number of model calls being 52. The model evidence is estimated by TBQ as 7.8739×10^{-4} , with the CI being $[7.7709, 7.9799] \times 10^{-4}$, which is in good agreement with the reference estimate by TMCMC. Further, the posterior samples and marginal densities estimated by TBQ are reported in Fig. 19. One can see that these results match well with the reference solutions generated by TMCMC. It is also observed that the posterior density of each parameter is multimodal and sharply peaked. The TBQ algorithm is then repeated ten times. The resultant number of model calls and the estimate of the model evidence, both averaged across the ten replications, are 59.4 and 7.6478×10^{-4} respectively, and the CoV of the estimates across these ten runs is 3.81%. The above facts sufficiently demonstrated the high efficiency, accuracy and robustness of the TBQ for estimating both posteriors and model evidence suffered from multimodal and sharply peaked feature.

6.5. An airplane dynamic model with real measurements

Consider the dynamic model of a toy airplane, for which one can refer to Refs. [4,18] for detailed descriptions of the prototype, numerical simulator, experiments and measurements. Due to space constraints, all of this information will not be discussed in detail here. Basically, the target is to identify the values of the Young's modulus of five connector parts, denoted as $E_1 \sim E_5$, based on the measurements of the natural frequencies of the first five orders, which are given as 20.3694, 40.7582, 92.0218, 106.2751, and 143.9561, respectively. The prior distribution is assumed to be truncated normal with the same parameters reported in Ref. [18], and the likelihood is formulated following Eq. (2). For verifying the effectiveness of the TBQ algorithm to sharp posterior, the STDs of the measurement noise are assumed to be 1% of the measured values.

Initialized with $N_0 = 12$ samples, the TBQ automatically produces two intermediate tempering stages, with the tempering parameters being 0.1519 and 1 respectively, to approach the posterior density. It consumes a total number of 17 model calls. The results of posterior densities generated by TBQ and TMCMC are reported in Fig. 20. It is seen that the results generated by

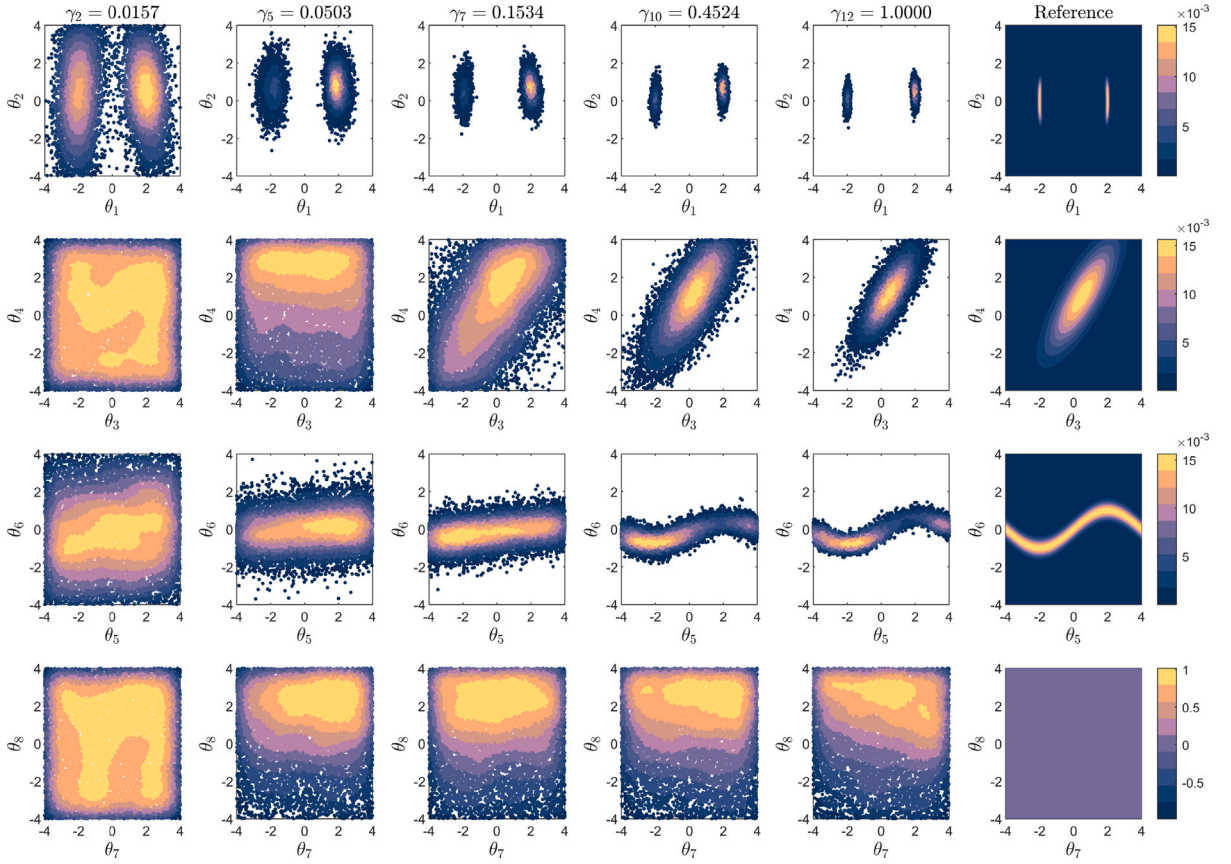


Fig. 15. Sample results of intermediate posterior densities for case 3 of the MDMM example, with the first five column being the samples for five stages selected from a total number of eleven tempering stages, and the last column being the true target density (unnormalized).

TBQ match well with those generated by TMCMC. These results show that, among the five parameters, only the fourth show big difference between prior and posterior, and all the other four parameters are non-influential, and thus their uncertainty have not be obviously reduced. One notes that the agreement of prior and posterior for a parameter does not mean that the prior is accurate, but instead, indicates it makes no contribution to the model response uncertainty. This has also been validated by the global sensitivity analysis results reported in Ref. [18]. In this case, the uncertainty of this parameter cannot be, also not necessary to be, reduced via BMU. This also inspires us that, for high-dimensional BMU problem, one can first perform the global sensitivity analysis to screen out the non-influential parameters, this way to reduce the dimension of the BMU problem. Additionally, The mean prediction of the model evidence by TBQ and TMCMC are 2.3143×10^{-5} and 2.3345×10^{-5} respectively, which match well with each other. The CI of Z generated by TBQ is $[2.3001, 2.3698] \times 10^{-5}$, indicating the high accuracy of TBQ.

6.6. Final remarks

As final remarks, I first revisit the topic of algorithm parameter settings, as this is generally the primary concern of users. Basically, the inputs to TBQ include three sample size parameters N_0 , N_{MC} and N_{est} , the confidence level parameter α , the stopping threshold ϵ_α , and ζ for determining the divergence of the posteriors of two consecutive tempering stages. For all the numerical experiments presented in Section 6, N_0 and N_{MC} are set as 12 and 10^4 respectively, and results in Figs. 6, 11 and 14 all demonstrated the robustness of this setting. Intuitively, N_0 should be pre-specified to ensure that a valid GP model can be initially achieved, and it can also be set to twice or thrice the parameter dimension. N_{MC} should be sufficiently large to ensure that the variation of the MC estimator for Eq. (25) is sufficiently small, and it is found that, given ζ not exceeding 3, setting N_{MC} as 10^4 is sufficient. N_{est} , the size of the sample matrix S , is utilized for estimating the model evidence, given the stopping condition being satisfied. It can be set as sufficiently large, e.g., 10^7 , as the integrand admits a closed-form expression, and only one computation is required for each stage. The values of α and ϵ_α should be specified in pairs. Intuitively, the value of α provide a balance between exploration and exploitation. Given smaller value of α , higher value of β can be achieved, and the algorithm is more likely to be in favor of exploration. However, results in Section 6 show that, given recommended value of α as 0.05~0.6, the performance of the algorithm is not sensitive to its value. Additionally, smaller value of α results in larger CI, and thus higher value of ϵ_α should be specified.

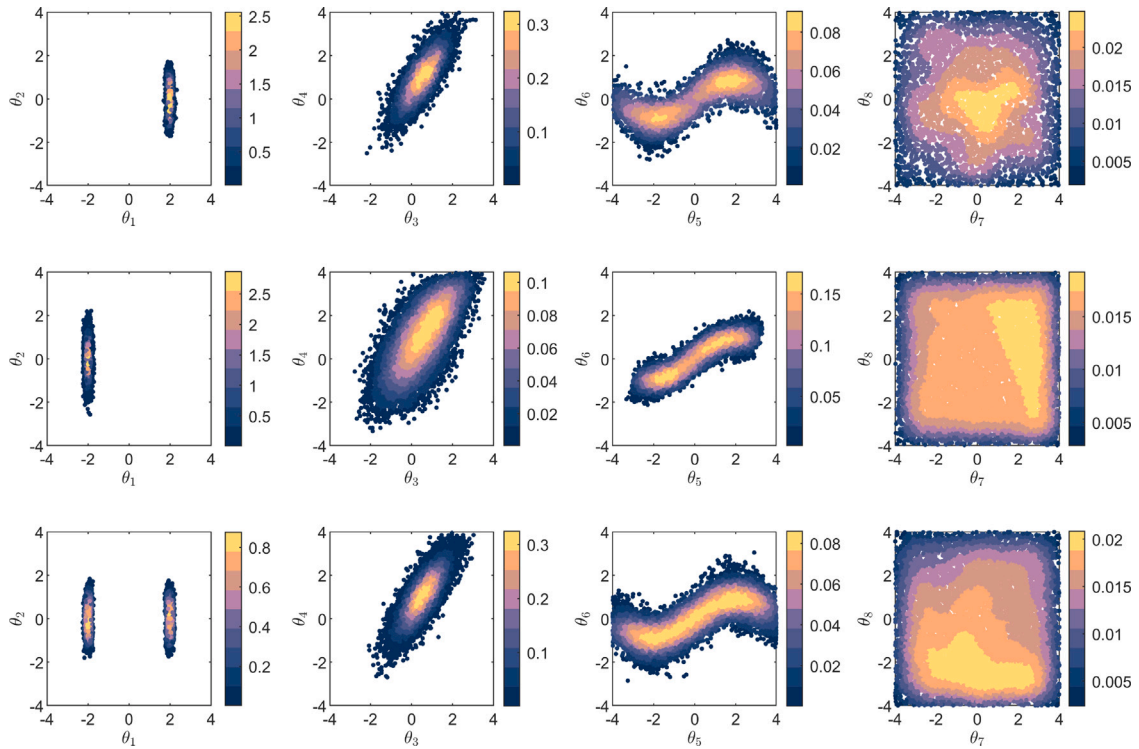


Fig. 16. Posterior samples generated by HMC (first row), VBQ (second row) and TCMCM (third row) for case 3 of the MDMM example.

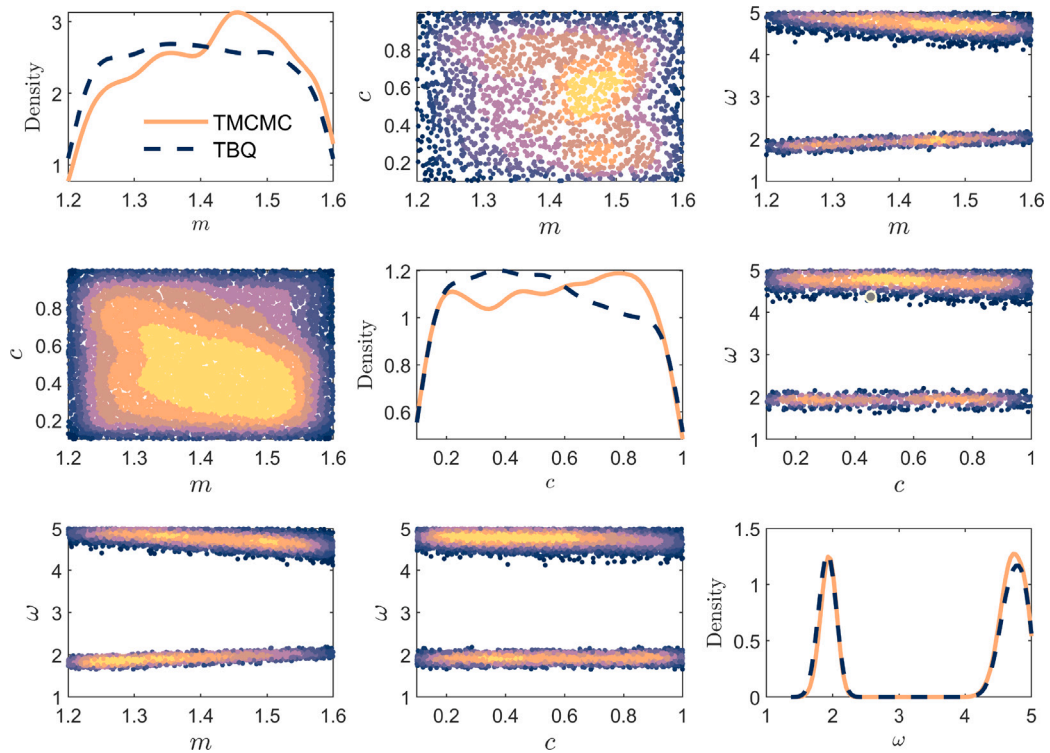


Fig. 17. Results of posterior densities for the nonlinear dynamic models, where the three sub-graphs in the lower triangle display joint posterior samples for each parameter pair, and those in the upper triangle present reference results generated by TCMCM. The three diagonal sub-graphs compare marginal posterior densities, produced via kernel density estimation from samples generated by TCMCM and TBQ.

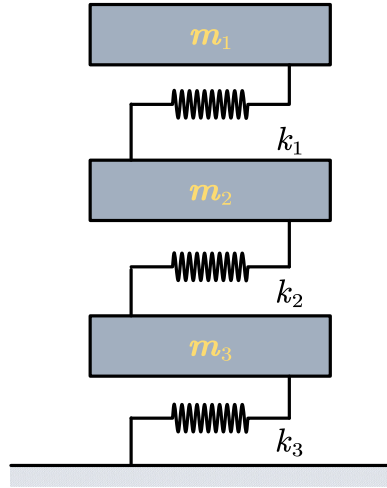


Fig. 18. Illustration of the 3DoF system.

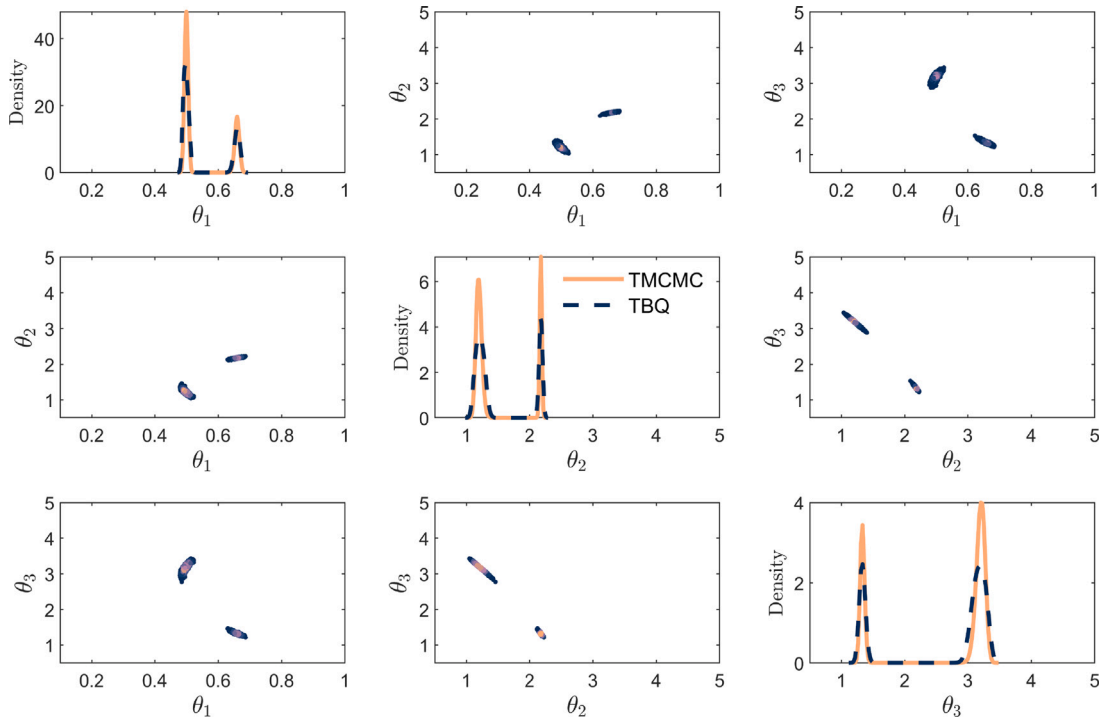


Fig. 19. Results of the posterior for the 3DoF system, where the three diagonal sub-graphs refer to the marginal posterior densities generated with kernel density estimation based on the posterior samples generated with TCMC and BQ, the sub-graphs in the upper triangle refer to the reference joint posterior samples generated using TCMC, and those in the lower triangle indicate those generated by TBQ.

Users can pre-specify its value following my suggestion, that is to set it as 0.06~0.3 for intermediate stages and 0.02~0.1 for the last stage, but the specific values should be given based on the above intuition. The last parameter ζ determines the magnitude of the divergence between the posterior densities of two consecutive stages, and higher value results in higher divergence and then less intermediate stages. For all the examples in Section 6, setting ζ as any value between 1 and 2 works out, but the required number of model calls to achieve the same stopping condition may differ. Results in Fig. 7 indicates that the performance of TBQ is relatively robust with respect to the value of ζ . The users can choose a relatively small value, say 1, if the posterior is not extremely divergent from the prior; otherwise, a higher value is recommended, but should be better not to exceed 2, although sometimes it even works for $\zeta = 3$.

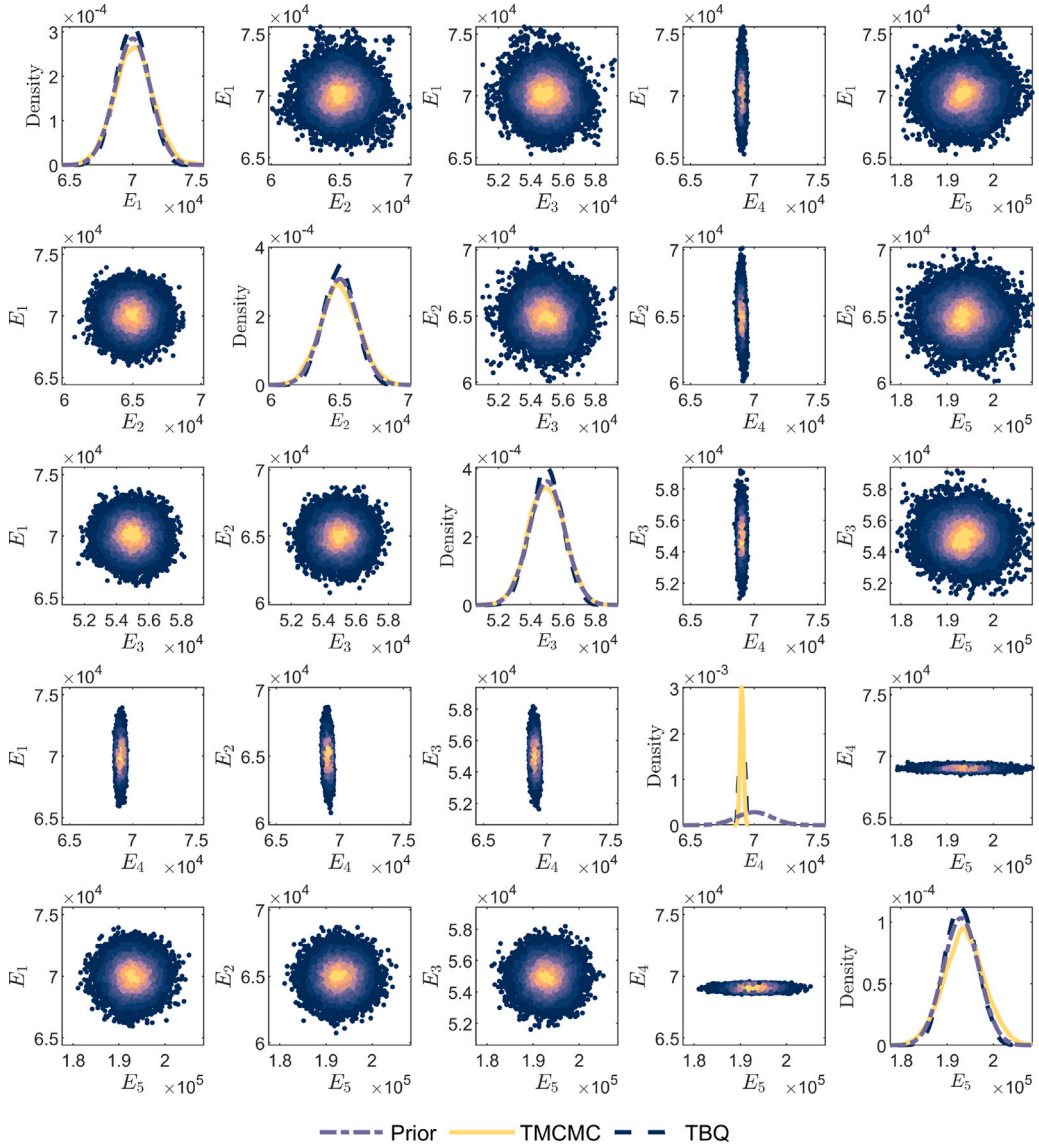


Fig. 20. Results for the airplane dynamic models, with the lower triangle subplots being generated by TBQ, and upper triangle subplots being generated by TCMC. The diagonal subplots show the marginals of both prior and posterior densities.

As explained in Section 5.3, it is suggested to estimate each intermediate model evidence Z_j using MCS. This is generally practical as the integrand (posterior mean of likelihood) admits a closed-form expression and the estimation needs to be implemented only once for each tempering stage. One can also use other schemes for estimation. For example, in case Z take very large or small values, one can estimate the logarithm of Z , instead of Z directly, using e.g., Chib's method [50]; purely based on the posterior samples produced by TBQ, one can estimate Z using the harmonic mean estimator [51]; or based on MCMC samples producing all tempering stages, one can estimate Z based on the products of the evidence ratios Z_i/Z_{j-1} , as suggested in TCMC [50]. One can refer to Ref. [9] for an overview of the available estimation schemes.

Compared with TCMC, TBQ also show other favorable features, except being much more efficient. In TCMC, ς is usually recommended to be 1, but in TBQ, it can be much more flexible to allow for higher transitional efficiency from prior to posterior. As show in the five test examples, setting ς between 1 and 2 all produced accurate and robust estimation. Another feature is that the MCMC samples produced by TBQ show much less particle degradation, as revealed by Fig. 17. Both features are due to the extremely high efficiency of generating MCMC samples in TBQ, as calling the simulator is not required during this process.

Definitely, any performance improvements come with a pain. Compared with the sampling methods, the proposed methods are less effective to high-dimensional problems due to kernel degradation—given the same distance and scale parameter in each dimension, a kernel tends to decrease with the increment of dimension. This is perhaps the limitation of all kernel methods [52].

Some schemes are available to alleviate this limitation, which include, for example, additive GP model which builds high-dimensional kernel with low-dimensional kernels [53], the dimension reduction techniques [52], and the Global Sensitivity Analysis (GSA) which aims at screening out the non-influential parameters before implementing TBQ (see Ref. [54] for a comprehensive review of the available GSA methods). Further more, the transitional learning strategy in this work also provides a possible approach for addressing high-dimensional problems. During the early tempering stage where γ is small, the main target is to explore the whole support of the prior to identify the location of modes, and in this case, one can restrict the minimum value of the kernel's scale parameter, with the aim of alleviating kernel degradation and enhancing the algorithm's exploration performance; while for later stage as γ increases, the scale parameters can be reduced to increase the exploitation of local modes, and meanwhile the sparse GP model can be used to balance the exploration of multiple modes and the exploitation of each single modes. Following this road-map, it is expected to extend the TBQ method to high-dimensional scenarios in the future.

Beyond TBQ, there have been a number of other developments for improving TMCMC, such as the X-TMCMC which aims at improving TMCMC with adaptive Kriging surrogate models and Langevin adjusted proposals [55], and the Bayesian pattern search which consists of a first step for deterministic pattern search, and then a second step of using TMCMC for quantifying the influence of the prevailing uncertainties associated with each mode [56]. TBQ employs the exponential-impact-inspired BQ scheme to improve TMCMC, with the aim of actively learning the tempering parameters and estimating the intermediate posteriors as well as the associated mode evidences with desired accuracy, and thus is completely distinct from the above-mentioned methods. Due to limit of space, a comprehensive comparison of the performance of TBQ with those methods will not be conducted, but interested readers can use the publicly code of this study to implement the examples reported in these references.

7. Conclusions

This paper has made substantial improvements to BQ method for addressing the common challenge in BMU problems, of which the posteriors are trickily behaved with multiple disconnected modes, highly sharp modes and nonlinear dependencies. The exponential impact which has the potential to boost the performance of the BQ methods has been primarily investigated. Then, this positive impact is leveraged for several key developments aiming at boosting the performance of the BQ method, including new scheme for quantifying prediction uncertainties, new acquisition functions for efficient and sequential design of training points, adaptive transitional scheme for learning sharp posteriors, elaborately designed estimators which avoids error accumulation across tempering stages, etc. All these meticulously designed components combine to form the robust and efficient TBQ algorithm for learning the complex posteriors and the associated model evidences. Results of the numerical experiments and engineering applications have highlighted the following features of TBQ.

- I. Benefiting from the exponential impact, the proposed acquisition functions are more likely to locate the training points within the support of true modes of (intermediate) posterior densities, as seen from Figs. 4, 5 and 9. This feature makes the acquisition functions like a high-performance engine of the proposed algorithm.
- II. As the error accumulation across tempering stages has been eliminated, the posterior and model evidence of each intermediate stage are not necessarily learned with high accuracy, only those of the last stage needs to be estimated with pre-specified low error tolerance. This relaxation significantly reduce the required number of model calls consumed by intermediate tempering stages.
- III. A greater span between consecutive tempering stages can be achieved by setting ζ as a higher value, instead of 1 as suggested by TMCMC. This is mainly benefited from the effectiveness of the proposed BQ for learning posteriors which are moderately far away from their priors, and further results in high transitional efficiency as less tempering stages are required. This is supported by the results of case 2 and 3 of the MDMM example as ζ is set as 1.8 to achieve high transitional efficiency.
- IV. Befitting from the proposed quantification of prediction uncertainties and the elimination of error accumulation, it is flexible to make a trade-off between accuracy and efficiency for estimating the posteriors and model evidences of the last tempering stage.

All the above features make the proposed TBQ algorithm remarkably suitable for addressing the BMU problems with tricky posteriors.

Despite the above features, I believe there are still potential for improvements. For example, designing more effective acquisition functions to achieve higher convergence rate, which may be achieved by, e.g., the idea of prospective learning [57]; extending the method to high-dimensional cases with hundreds of model parameters, which may be attained by introducing adaptive scheme for scaling the kernel; etc. All these will be investigated in future work.

The Matlab code for implementing the TBQ algorithm is available at <https://github.com/PengfeiWei-NWPU/TBQ-for-Bayesian-Model-Inference.git>.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Data availability

Data will be made available on request.

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